Pico Example

Assembly Design Document

1 Description

This example assembly is designed to run on the Raspberry Pi Pico. It includes a small collection of components that demonstrate how Adamant can be deployed onto a bare metal embedded system. In particular, the assembly demonstrates a simple rate group system, commanding, telemetry reporting using events, data products, and packets, and a simple fault detection and correction scheme. Explore the tables and diagrams below to learn more about the design of the example assembly.

2 Design

2.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the assembly.

- **Number of Components** - 26
- **Number of Component Types** - 23
- **Number of Active Components** - 7
- **Number of Passive Components** - 19
- **Number of Components with Queue** - 10
- **Number of Components without Queue** - 16
- **Number of Components with Events** - 21
- **Number of Components with Data Products** - 17
- **Number of Components with Data Dependencies** - *None*
- **Number of Components with Packets** - 11
- **Number of Components with Commands** - 17
- **Number of Components with Parameters** - 2
- **Number of Components with Faults** - 2
- **Number of Connections** - 134
- **Number of Events** - 149
- **Number of Data Products** - 38
- **Number of Data Dependencies** - *None*
- **Number of Packets** - 12
- **Number of Commands** - 55
- **Number of Parameters** - 6
- **Number of Faults** - 4

2.2 Components

All of the components in the Pico Example assembly are reusable components that can be found within the Adamant repository (in *src/components/*) with a few exceptions:

- **`Adc_Data_Collector`** - This component collects internal telemetry from the Raspberry Pi Pico including system voltage and temperature.
- **`Counter`** - This is an extremely simple component that produces an incrementing counter as telemetry.
- **`Fault_Producer`** - This is an component that can be used to inject a fault into the system via command.
- **`Oscillator`** - This is a simple component that produces two sinusoidal outputs into telemetry.

These specific component are located in the Example repository in *src/components/*. A full list of the components included in the assembly can be seen below. Note that components marked as active execute on their own task within the Ada runtime.

Table 1: Pico Example Components

Name	**Type**	**Has Queue**	**Execution**
`System_Time_Instance`	`Gps_Time`	`no`	`passive`
`Ticker_Instance`	`Ticker`	`no`	`active`
`Tick_Divider_Instance`	`Tick_Divider`	`no`	`passive`
`Slow_Rate_Group`	`Rate_Group`	`yes`	`active`
`Fast_Rate_Group`	`Rate_Group`	`yes`	`active`
`Watchdog_Rate_Group`	`Rate_Group`	`yes`	`active`
`Ccsds_Command_ Depacketizer_Instance`	`Ccsds_Command_ Depacketizer`	`no`	`passive`
`Command_Router_Instance`	`Command_Router`	`yes`	`active`
`Event_Filter_Instance`	`Event_Filter`	`no`	`passive`
`Event_Limiter_Instance`	`Event_Limiter`	`no`	`passive`
`Event_Packetizer_ Instance`	`Event_Packetizer`	`no`	`passive`
`Product_Database_ Instance`	`Product_Database`	`no`	`passive`
`Product_Packetizer_ Instance`	`Product_Packetizer`	`yes`	`passive`

Ccsds_Packetizer_ Instance	Ccsds_Packetizer	no	passive
Ccsds_Serial_Interface_ Instance	Ccsds_Serial_Interface	yes	active
Counter_Instance	Counter	yes	passive
Oscillator_A	Oscillator	yes	passive
Oscillator_B	Oscillator	yes	passive
Adc_Data_Collector_ Instance	Adc_Data_Collector	no	passive
Zero_Divider_Instance	Zero_Divider	no	passive
Task_Watchdog_Instance	Task_Watchdog	no	passive
Fault_Producer_Instance	Fault_Producer	no	passive
Fault_Correction_ Instance	Fault_Correction	yes	active
Cpu_Monitor_Instance	Cpu_Monitor	no	passive
Queue_Monitor_Instance	Queue_Monitor	no	passive
Stack_Monitor_Instance	Stack_Monitor	no	passive

Table 2: Pico Example Component Item Counts

Component Name	**Connectors**	**Commands**	**Events**	**Data Products**	**Data Dependencies**	**Parameters**	**Packets**	**Faults**
System_Time_ Instance	1	0	0	0	0	0	0	0
Ticker_Instance	2	0	0	0	0	0	0	0
Tick_Divider_ Instance	4	0	1	0	0	0	0	0
Slow_Rate_Group	6	0	5	1	0	0	0	0
Fast_Rate_Group	6	0	5	1	0	0	0	0
Watchdog_Rate_ Group	6	0	5	1	0	0	0	0
Ccsds_Command_ Depacketizer_ Instance	8	1	7	2	0	0	1	0

Command_Router_ Instance	10	4	18	7	0	0	0	0
Event_Filter_ Instance	9	7	12	3	0	0	1	0
Event_Limiter_ Instance	9	8	14	3	0	0	1	0
Event_ Packetizer_ Instance	7	1	0	2	0	0	1	0
Product_ Database_ Instance	8	5	18	2	0	0	1	0
Product_ Packetizer_ Instance	7	5	10	0	0	0	2	0
Ccsds_ Packetizer_ Instance	2	0	0	0	0	0	0	0
Ccsds_Serial_ Interface_ Instance	4	0	3	0	0	0	0	0
Counter_Instance	6	3	6	0	0	0	1	0
Oscillator_A	7	3	6	1	0	3	0	0
Oscillator_B	7	3	6	1	0	3	0	0
Adc_Data_ Collector_ Instance	3	0	0	3	0	0	0	0
Zero_Divider_ Instance	4	1	3	0	0	0	1	0
Task_Watchdog_ Instance	9	4	10	4	0	0	0	2
Fault_Producer_ Instance	5	2	3	0	0	0	0	2
Fault_ Correction_ Instance	7	5	11	4	0	0	0	0
Cpu_Monitor_ Instance	7	1	2	1	0	0	1	0
Queue_Monitor_ Instance	7	1	2	1	0	0	1	0
Stack_Monitor_ Instance	7	1	2	1	0	0	1	0

Component Descriptions:

- **`System_Time_Instance`** - This component provides the system time for the assembly.
- **`Ticker_Instance`** - This component uses a periodic signal to drive the assembly rate groups.
- **`Tick_Divider_Instance`** - This component divides a periodic signal into intervals suitable for the assembly rate groups.
- **`Slow_Rate_Group`** - This component provides a 0.5 Hz task for other components to execute on periodically.
- **`Fast_Rate_Group`** - This component provides a 5 Hz task for other components to execute on periodically.

- **`Watchdog_Rate_Group`** - This component provides a 1 Hz task for the watchdog components to execute on periodically.
- **`Ccsds_Command_Depacketizer_Instance`** - This component converts CCSDS packets containing commands to valid Adamant formatted command types.
- **`Command_Router_Instance`** - This component provides command and command response routing throughout the assembly.
- **`Event_Filter_Instance`** - This component filters events by ID.
- **`Event_Limiter_Instance`** - This component filters out events that spam the system too frequently.
- **`Event_Packetizer_Instance`** - This component gathers events and packetizes them for downlink.
- **`Product_Database_Instance`** - This component serves as the database for data products throughout the system.
- **`Product_Packetizer_Instance`** - This component periodically fetches values from the Product Database and packetizes them for downlink.
- **`Ccsds_Packetizer_Instance`** - This component converts Adamant formatted packets to CCSDS for downlink.
- **`Ccsds_Serial_Interface_Instance`** - This component manages the Raspberry Pi Pico UART interface for command uplink and telemetry downlink.
- **`Counter_Instance`** - This component periodically produces a count data product.
- **`Oscillator_A`** - This component periodically produces an oscillating data product.
- **`Oscillator_B`** - This component periodically produces an oscillating data product.
- **`Adc_Data_Collector_Instance`** - This component collects data from the internal Raspberry Pi Pico analog to digital converted (ADC) and reports it as telemetry.
- **`Zero_Divider_Instance`** - This component responds to a command that divides by zero if received. This can be used to trigger a fault condition within the processor.
- **`Task_Watchdog_Instance`** - This component monitors other critical active components (tasks) within the assembly to make sure they continue to run.
- **`Fault_Producer_Instance`** - This component can be used to induce a fault into the system by command.
- **`Fault_Correction_Instance`** - This component produces a corrective action (a command) for any fault that is thrown in the system.
- **`Cpu_Monitor_Instance`** - This component produces a packet that includes the CPU usage percentage for each task and interrupt in the system.
- **`Queue_Monitor_Instance`** - This component produces a packet that includes the current and maximum queue usage for each component in the system.
- **`Stack_Monitor_Instance`** - This component produces a packet that includes the maximum stack and secondary stack usage for each component in the system.

2.3 Views

This section shows the example assembly visually as a set of *views*. Each shows a specific set of components and connections (while not showing other components and connections) in order to highlight a particular function of the assembly. Components that are bold are *active*, meaning they have an Ada task assigned to them. Connections are labeled with the type that is passed along them. A dotted line indicates that the connection is asynchronous, meaning the data is put onto a queue for later processing. A solid line indicates that the connection is synchronous, meaning processing of that data occurs right when the data is passed along the connector.

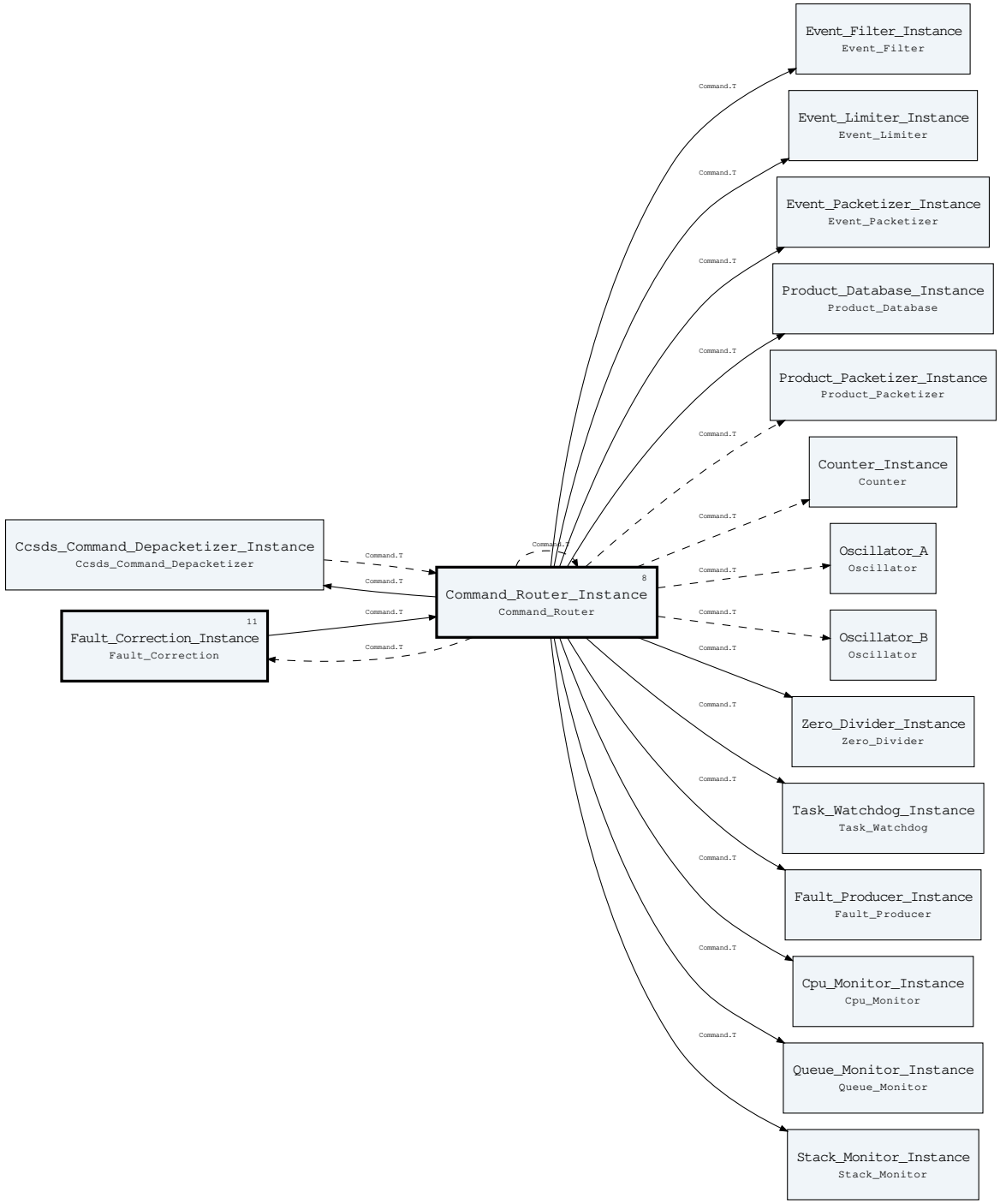

Figure 1: **Command View:** This view shows how commands are routed throughout the assembly. Commands are contained in the Command.T data type. Commands coming from the ground system originate from the Ccsds_Command_Depacketizer_Instance and then get passed the Command_Router_Instance. The router looks at the command's ID, determines the destination component for which the command is intended, and then forwards the command to that appropriate destination component. When a destination component receives a command, it will execute it and pass a Command_Response.T data type back to the command router (shown in the Command Response View). Note that the Fault_Correction_Instance can also produce commands in order to correct a system fault. Commands from the Fault_Correction_Instance are passed to the router synchronously, bypassing the standard command queue that the Ccsds_Command_Depacketizer_Instance uses.

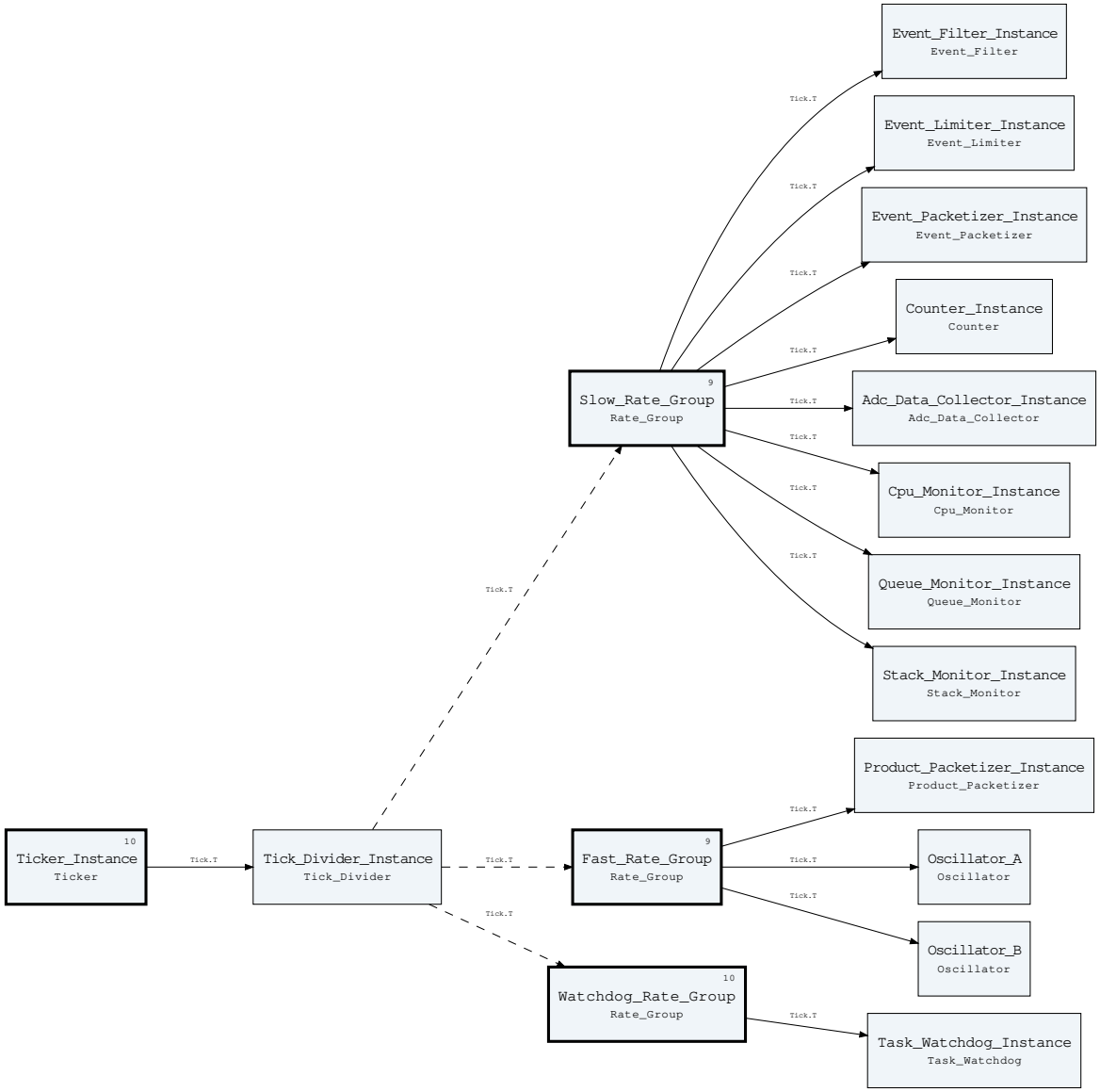

Figure 2: **Tick View:** In this assembly, Tick.T types are used to indicate when periodic tasks should be run. This view shows the rate groups that exist in the assembly. The Ticker_Instance runs in a high priority task and produces a Tick.T at 5 Hz. This tick is forwarded to the Tick_Divider_Instance which takes the 5 Hz signal and divides it up among 3 different rate groups. The Slow_Rate_Group runs at 0.5 Hz. The Fast_Rate_Group runs a 5 Hz. The Watchdog_Rate_Group runs as 1 Hz. Each of these rate group components will call the components connected to them every time they receive a tick from upstream. This architecture allows for periodic execution of certain functions, such as collecting telemetry or creating packets. Note that the actual execution of each rate group is also interleaved by the Ada scheduler. Each rate group is an active component, which executes on a task with a priority shown in the top right hand corner of the component. Rate groups produce telemetry relating how long it takes them to execute, both in CPU time and in wall clock time.

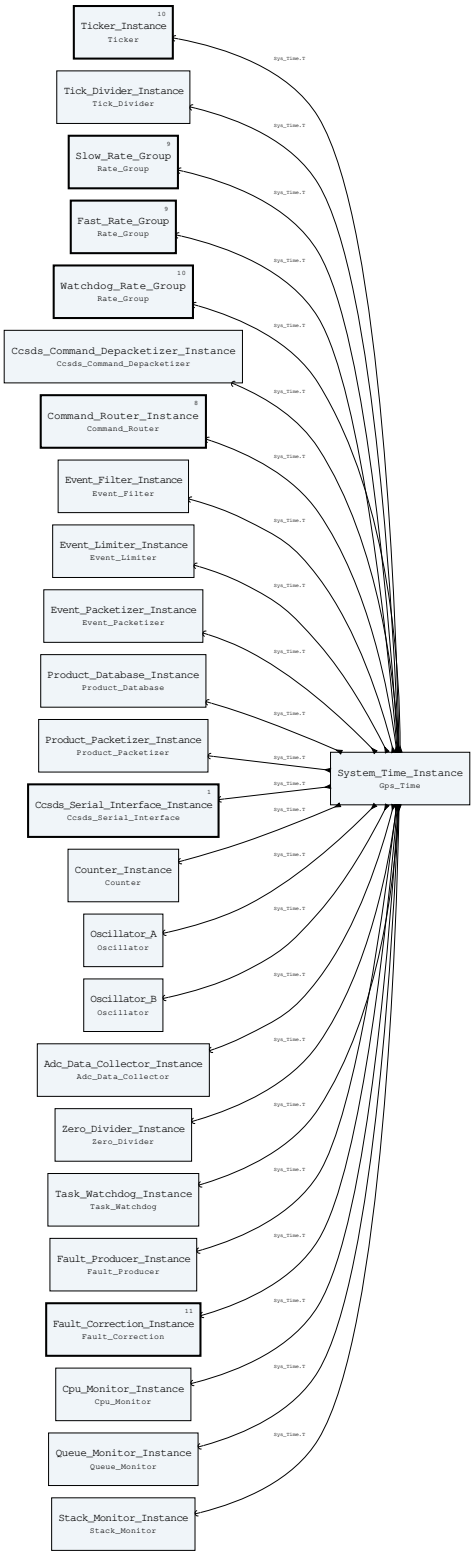

Figure 3: **Time View:** All components in the system fetch time by requesting it from the System_Time_Instance. By implementing the time system at the Adamant architectural level, different time sources and time synchronization schemes can be easily swapped into the system by replacing System_Time_Instance with a more tailored version. Note that synchronous connectors exhibit extremely low overhead, so exposing anything at the architectural level should not be prohibitive in terms of performance.

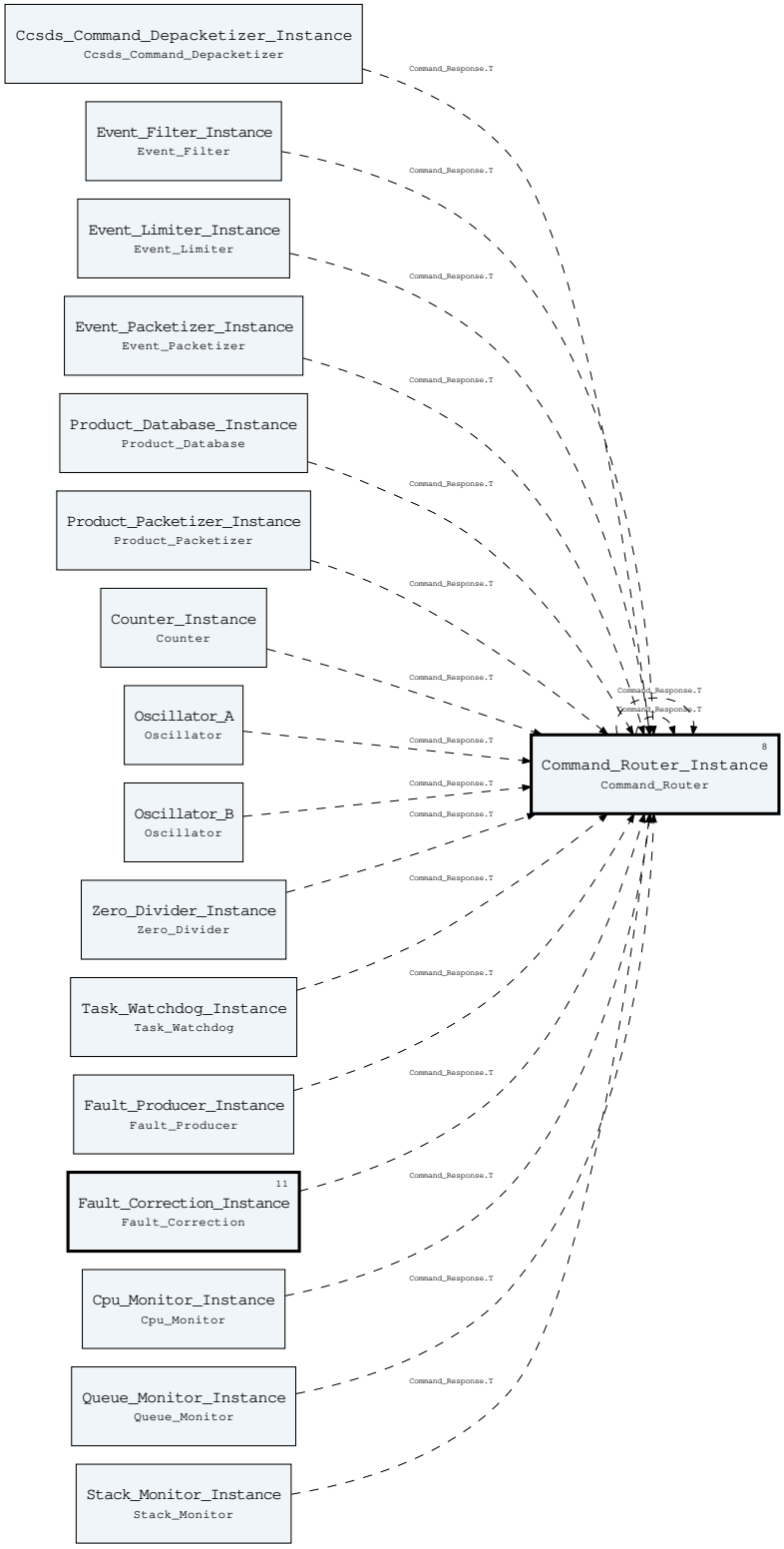

Figure 4: **Command Response View:** After a component executes a command received from the Command_Router_Instance, it passes back a Command_Response.T type to the router letting it know if the command succeeded or failed. This connection is also used to register all the commands with the router an initialization. This allows the Command_Router_Instance to create the internal routing table that it uses to route incoming commands.

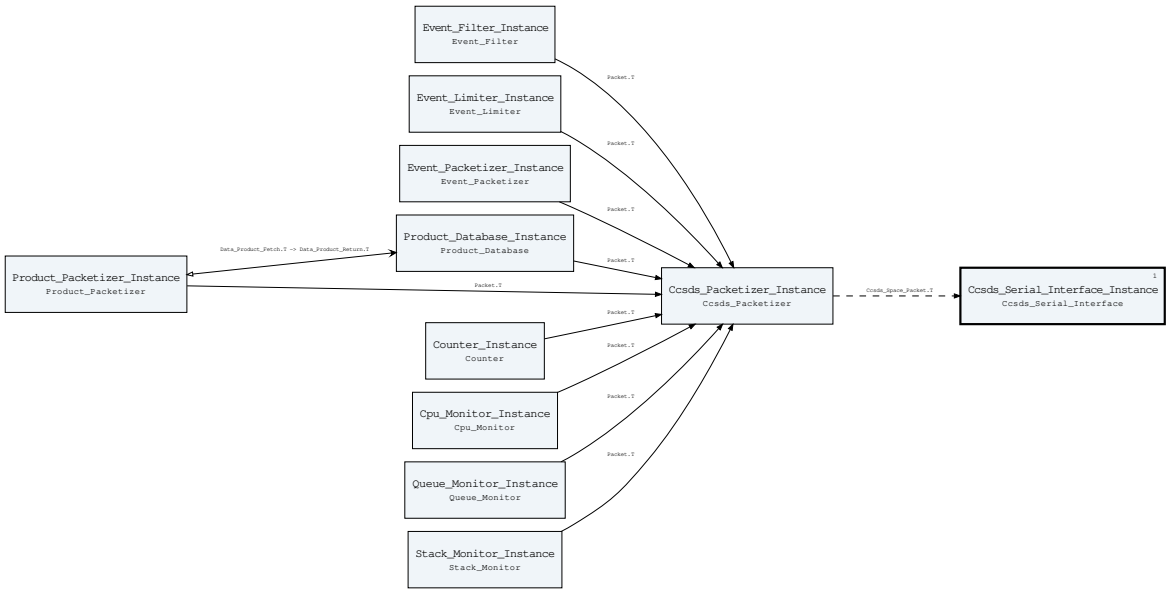

Figure 5: **Downlink View:** Any packet produced by the assembly is sent to the Ccsds_Packetizer_Instance. This component takes the Adamant formatted packets, converts them to CCSDS, and sends them asynchronously to the Ccsds_Serial_Interface_Instance. The serial interface component will take packets stored on its queue and transmit them over the Raspberry Pi Pico UART. The Product_Packetizer_Instance periodically requests data products from the Product_Database_Instance to create packets containing data from multiple components. These packets are also forwarded to the Ccsds_Packetizer_Instance for downlink.

Figure 6: **Uplink View:** The Ccsds_Serial_Interface_Instance receives data on the Raspberry Pi Pico UART. It is actively looking for a sync pattern followed by a CCSDS packet header. If it receives a valid CCSDS packet it forwards it along to the Ccsds_Commmand_Depacketizer_Instance. This component looks for commands in the CCSDS packet, extracts them, and then forwards them along the Command_Router_Instance for routing and later execution.

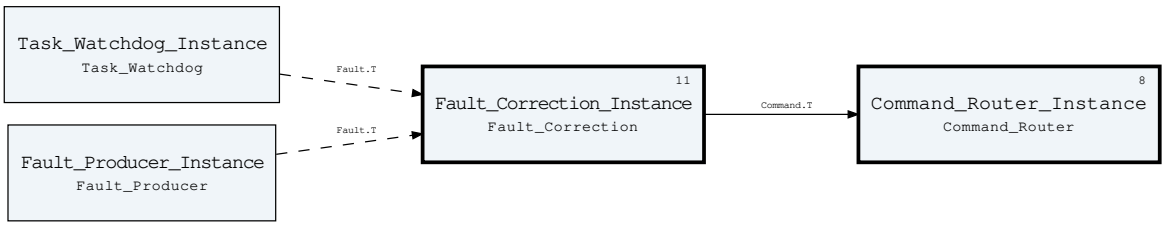

Figure 7: **Faults View:** Any component that produces faults will send them to the Fault_Correction_Instance. This component maps the fault to a response action. This action is in the form of a correction command which is sent to the Command_Router_Instance synchronously for execution. In the Raspberry Pi Pico assembly, two components can throw faults. The first is the Task_Watchdog_Instance. This component monitors some critical tasks to ensure that they are always running in a timely fashion. If one stops executing, a fault is thrown. The Fault_Producer_Instance is a simple component that throws a fault when commanded to. This can be used to inject a fault into the system for testing purposes.

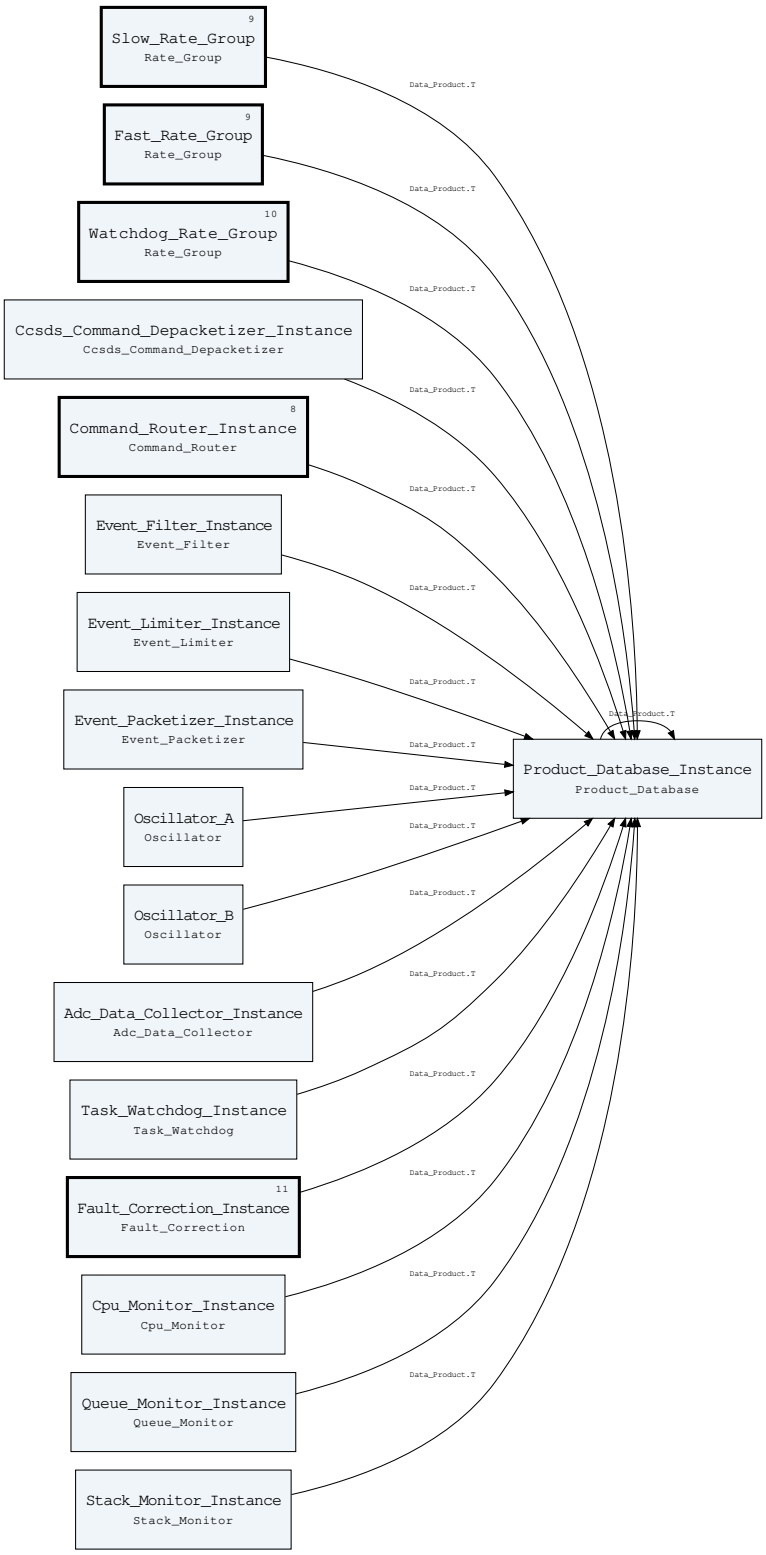

Figure 8: **Data Products View:** Any data product produced by a component during execution is sent synchronously to on onboard storage database, the Product_Database_Instance. This component stores the latest value and timestamp of each data product in the system. Other components, such as the Product_Packetizer_Instance can then fetch these data products at a later time for limit checking, packetization, etc.

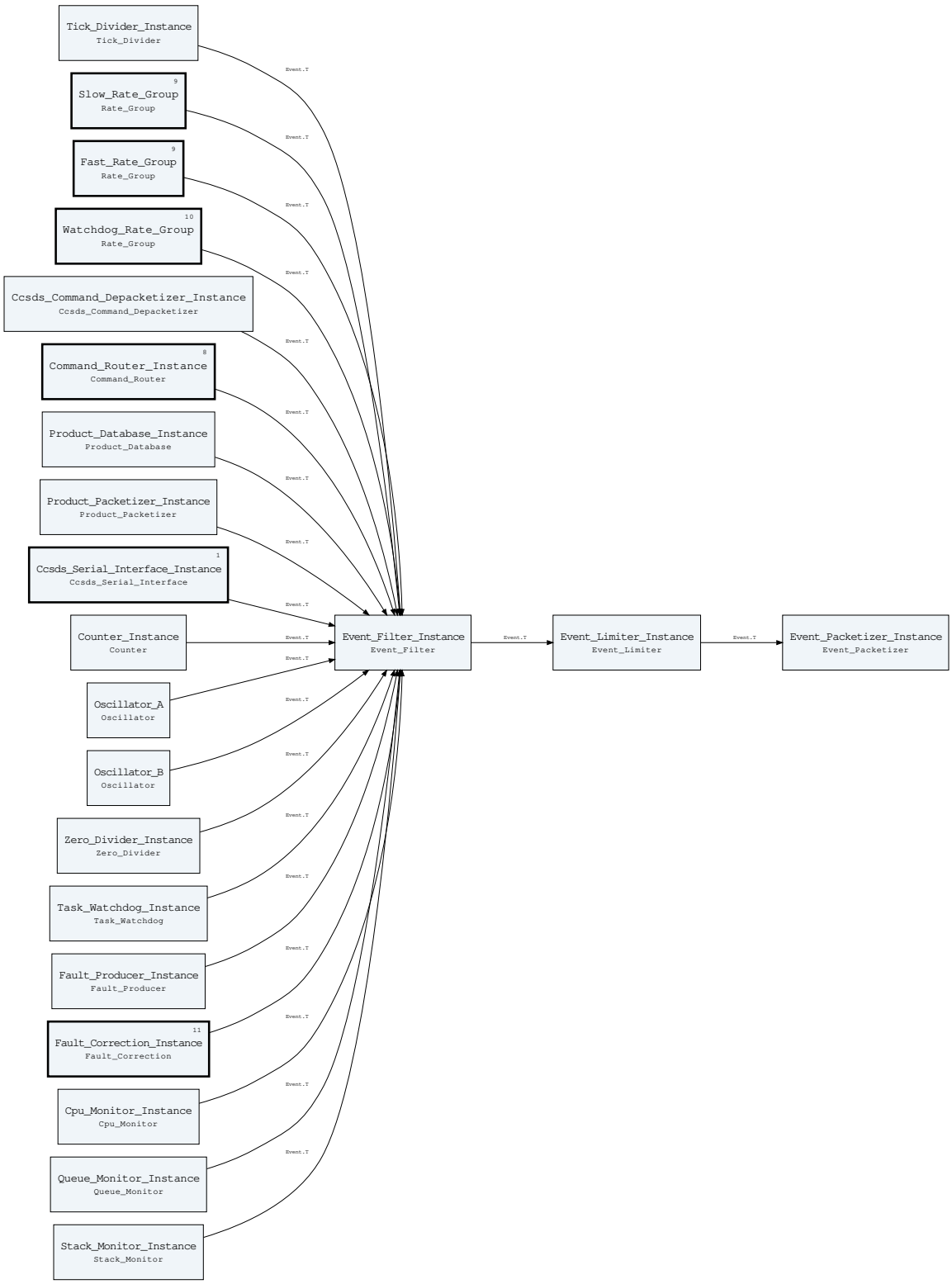

Figure 9: **Events View:** Events are produced by components when something interesting happens. All components send their events first to the Event_Filter_Instance. This component can enabled/disable individual events by ID. Next, events are forwarded to the Event_Limiter_Instance. This component will start limiting problematic spam events that might flood the system. Any event that passes both of these filtering components is passed to the Event_Packetizer_Instance for collection in a packet for later downlink.

2.4 Task Priorities

The table below outlines the system tasks for the Pico Example assembly. Task names are in the form *component_name.task_name*. The priority *rank* is a number from 1 to n denoting how the priority of a component's task compares to others in the system. A rank of 1 is the highest priority in the system. The *priority value* is the actual priority number provided to the system scheduler. A larger *priority value* value signifies a higher priority task.

Table 3: Pico Example Component Task Priorities

Task Number	**Task Name**	**Priority Rank**	**Priority Value**
0	`Fault_Correction_Instance.Active_Task`	1	11
1	`Ticker_Instance.Active_Task`	2	10
2	`Watchdog_Rate_Group.Active_Task`	2	10
3	`Slow_Rate_Group.Active_Task`	3	9
4	`Fast_Rate_Group.Active_Task`	3	9
5	`Command_Router_Instance.Active_Task`	4	8
6	`Ccsds_Serial_Interface_Instance.Active_Task`	5	1
7	`Ccsds_Serial_Interface_Instance.Listener_Task`	6	0

2.5 Commands

The table below shows the commands for the Pico Example assembly.

Table 4: Pico Example Commands

Command ID	**Command Name**	**Argument Type**
`0x0001 (1)`	`Ccsds_Command_Depacketizer_Instance.Reset_Counts`	-
`0x0002 (2)`	`Command_Router_Instance.Noop`	-
`0x0003 (3)`	`Command_Router_Instance.Noop_Arg`	`Command_Router_Arg.T`
`0x0004 (4)`	`Command_Router_Instance.Noop_Response`	-

0x0005 (5)	Command_Router_Instance. Reset_Data_Products	-
0x0006 (6)	Event_Filter_Instance. Filter_Event	Event_Filter_Single_Event_ Cmd_Type.T
0x0007 (7)	Event_Filter_Instance. Unfilter_Event	Event_Filter_Single_Event_ Cmd_Type.T
0x0008 (8)	Event_Filter_Instance. Filter_Event_Range	Event_Filter_Id_Range.T
0x0009 (9)	Event_Filter_Instance. Unfilter_Event_Range	Event_Filter_Id_Range.T
0x000a (10)	Event_Filter_Instance. Enable_Event_Filtering	-
0x000b (11)	Event_Filter_Instance. Disable_Event_Filtering	-
0x000c (12)	Event_Filter_Instance.Dump_ Event_States	-
0x000d (13)	Event_Limiter_Instance. Enable_Event_Limit	Event_Single_State_Cmd_ Type.T
0x000e (14)	Event_Limiter_Instance. Disable_Event_Limit	Event_Single_State_Cmd_ Type.T
0x000f (15)	Event_Limiter_Instance. Enable_Event_Limit_Range	Event_Limiter_Id_Range.T
0x0010 (16)	Event_Limiter_Instance. Disable_Event_Limit_Range	Event_Limiter_Id_Range.T
0x0011 (17)	Event_Limiter_Instance. Enable_Event_Limiting	-
0x0012 (18)	Event_Limiter_Instance. Disable_Event_Limiting	-
0x0013 (19)	Event_Limiter_Instance.Set_ Event_Limit_Persistence	Event_Limiter_Persistence_ Type.T
0x0014 (20)	Event_Limiter_Instance. Dump_Event_States	-
0x0015 (21)	Event_Packetizer_Instance. Send_Packet	-

0x0016 (22)	Product_Database_Instance. Clear_Override	Data_Product_Id.T
0x0017 (23)	Product_Database_Instance. Clear_Override_For_All	-
0x0018 (24)	Product_Database_Instance. Override	Data_Product.T
0x0019 (25)	Product_Database_Instance. Dump	Data_Product_Id.T
0x001a (26)	Product_Database_Instance. Dump_Poly_Type	Data_Product_Poly_Extract.T
0x001b (27)	Product_Packetizer_ Instance.Set_Packet_Period	Packet_Period.T
0x001c (28)	Product_Packetizer_ Instance.Enable_Packet	Packet_Id.T
0x001d (29)	Product_Packetizer_ Instance.Disable_Packet	Packet_Id.T
0x001e (30)	Product_Packetizer_ Instance.Send_Packet	Packet_Id.T
0x001f (31)	Product_Packetizer_ Instance.Enable_Packet_ On_Change	Packet_Id.T
0x0020 (32)	Oscillator_A.Set_Frequency	Packed_F32.T
0x0021 (33)	Oscillator_A.Set_Amplitude	Packed_F32.T
0x0022 (34)	Oscillator_A.Set_Offset	Packed_F32.T
0x0023 (35)	Oscillator_B.Set_Frequency	Packed_F32.T
0x0024 (36)	Oscillator_B.Set_Amplitude	Packed_F32.T
0x0025 (37)	Oscillator_B.Set_Offset	Packed_F32.T
0x0026 (38)	Zero_Divider_Instance. Divide_By_Zero	Packed_U32.T
0x0027 (39)	Task_Watchdog_Instance. Enable_Watchdog_Pet_Checks	-

0x0028 (40)	Task_Watchdog_Instance. Disable_Watchdog_Pet_Checks	-
0x0029 (41)	Task_Watchdog_Instance.Set_ Watchdog_Limit	Pico_Example_Task_Watchdog_ List_Watchdog_Limit_Cmd.T
0x002a (42)	Task_Watchdog_Instance.Set_ Watchdog_Action	Pico_Example_Task_Watchdog_ List_Watchdog_Action_Cmd.T
0x002b (43)	Fault_Producer_Instance. Throw_Fault_1	-
0x002c (44)	Fault_Producer_Instance. Throw_Fault_2	-
0x002d (45)	Fault_Correction_Instance. Enable_Fault_Response	Pico_Example_Fault_ Responses_Packed_Id_Type.T
0x002e (46)	Fault_Correction_Instance. Disable_Fault_Response	Pico_Example_Fault_ Responses_Packed_Id_Type.T
0x002f (47)	Fault_Correction_Instance. Clear_Fault_Response	Pico_Example_Fault_ Responses_Packed_Id_Type.T
0x0030 (48)	Fault_Correction_Instance. Clear_All_Fault_Responses	-
0x0031 (49)	Fault_Correction_Instance. Reset_Data_Products	-
0x0032 (50)	Cpu_Monitor_Instance.Set_ Packet_Period	Packed_U16.T
0x0033 (51)	Queue_Monitor_Instance.Set_ Packet_Period	Packed_U16.T
0x0034 (52)	Stack_Monitor_Instance.Set_ Packet_Period	Packed_U16.T
0x0063 (99)	Counter_Instance.Set_Count	Packed_U32.T
0x0064 (100)	Counter_Instance.Reset_ Count	-
0x0065 (101)	Counter_Instance.Set_Count_ Add	Operands.T

Command Descriptions:

- **`Ccsds_Command_Depacketizer_Instance.Reset_Counts`** - This command resets the internal counts for the data products.

- **`Command_Router_Instance.Noop`** - Simple NOOP command which produces an event saying that it was triggered. This can be used to self test the command routing system and verify system aliveness.
- **`Command_Router_Instance.Noop_Arg`** - Simple NOOP command which produces an event saying that it was triggered with a certain Arg. This can be used to self test the command argument validation system. Sending a command with an Arg value of 868 will cause the component to Fail the command. Any other value will produce a successfully executed command.
- **`Command_Router_Instance.Noop_Response`** - A NOOP command which self tests the command response forwarding mechanism. The command handler itself acts as a command sender component, and sends out a NOOP command with a registered Source Id. The Command Router should then send out an event saying that the command response was forwarded and received again by the Command Router.
- **`Command_Router_Instance.Reset_Data_Products`** - This command resets the values of all the component's data product to the values at initialization.
- **`Event_Filter_Instance.Filter_Event`** - Enable the event filtering for a specific event ID.
- **`Event_Filter_Instance.Unfilter_Event`** - Disable the event filtering for a specific event ID.
- **`Event_Filter_Instance.Filter_Event_Range`** - Enable the event filtering for a specific range of event IDs.
- **`Event_Filter_Instance.Unfilter_Event_Range`** - Disable the event filtering for a specific range of event IDs.
- **`Event_Filter_Instance.Enable_Event_Filtering`** - Enable the component to filter events that have been set to be filtered.
- **`Event_Filter_Instance.Disable_Event_Filtering`** - Disable the component so that all events will not be filtered. The event states will be maintained for when re-enabled.
- **`Event_Filter_Instance.Dump_Event_States`** - Dump a packet for the state of all events pertaining to if they are filtered or not.
- **`Event_Limiter_Instance.Enable_Event_Limit`** - Enable the event limiter for a specific event ID.
- **`Event_Limiter_Instance.Disable_Event_Limit`** - Disable the event limiter for a specific event ID.
- **`Event_Limiter_Instance.Enable_Event_Limit_Range`** - Enable the event limiter for a specific range of event IDs.
- **`Event_Limiter_Instance.Disable_Event_Limit_Range`** - Disable the event limiter for a specific range of event IDs.
- **`Event_Limiter_Instance.Enable_Event_Limiting`** - Enable the component to limit events that have been set to be limited.
- **`Event_Limiter_Instance.Disable_Event_Limiting`** - Disable the component so that all events will not be limited. The event states and counts will be maintained for when re-enabled.
- **`Event_Limiter_Instance.Set_Event_Limit_Persistence`** - Change the persistence of the event limiter for all events that are limited. Value must be between 1 and 7.
- **`Event_Limiter_Instance.Dump_Event_States`** - Dump a packet for the state of all events on if they are limited or not.
- **`Event_Packetizer_Instance.Send_Packet`** - Send a packet out next tick, unless there are no events stored within the component.
- **`Product_Database_Instance.Clear_Override`** - Clear the override condition for the data product of the provided ID.

- **`Product_Database_Instance.Clear_Override_For_All`** - Clear the override condition for all data products in the product store.
- **`Product_Database_Instance.Override`** - Override the value of a data product in the data product store. The value of this data product will be fixed to the commanded value, ignoring all other updates, until the override is cleared.
- **`Product_Database_Instance.Dump`** - Dump the data product of the provided ID in a packet.
- **`Product_Database_Instance.Dump_Poly_Type`** - Dump the data product of the provided ID into a poly type based on the provided offset and length.
- **`Product_Packetizer_Instance.Set_Packet_Period`** - Command to change the period of packet generation for a given packet id.
- **`Product_Packetizer_Instance.Enable_Packet`** - Command to enable the emission of a packet from the packetizer.
- **`Product_Packetizer_Instance.Disable_Packet`** - Command to disable the emission of a packet from the packetizer.
- **`Product_Packetizer_Instance.Send_Packet`** - Command to build specific packet and send it out on the next available tick. The packet is built and sent regardless of the packet being enabled or disabled.
- **`Product_Packetizer_Instance.Enable_Packet_On_Change`** - Command to enable the emission of a packet from the packetizer only when data products have changed since the last emission.
- **`Oscillator_A.Set_Frequency`** - Set the frequency of the oscillator in Hz
- **`Oscillator_A.Set_Amplitude`** - Set the amplitude of the oscillator
- **`Oscillator_A.Set_Offset`** - Set the Y offset of the oscillator
- **`Oscillator_B.Set_Frequency`** - Set the frequency of the oscillator in Hz
- **`Oscillator_B.Set_Amplitude`** - Set the amplitude of the oscillator
- **`Oscillator_B.Set_Offset`** - Set the Y offset of the oscillator
- **`Zero_Divider_Instance.Divide_By_Zero`** - You must provide the correct magic number as argument to this command for it to be executed.
- **`Task_Watchdog_Instance.Enable_Watchdog_Pet_Checks`** - Command to enable the watchdog component to check all connected components for incoming pets.
- **`Task_Watchdog_Instance.Disable_Watchdog_Pet_Checks`** - Command to disable the watchdog component to check all connected components for incoming pets.
- **`Task_Watchdog_Instance.Set_Watchdog_Limit`** - Set the limit value for the watchdog given an index and the new index value.
- **`Task_Watchdog_Instance.Set_Watchdog_Action`** - Sets the action of a petter given the index of that petter and the updated action. Note that actions cannot be promoted to fault if they were not provided a fault id.
- **`Fault_Producer_Instance.Throw_Fault_1`** - Throw the first fault.
- **`Fault_Producer_Instance.Throw_Fault_2`** - Throw the second fault.
- **`Fault_Correction_Instance.Enable_Fault_Response`** - Enable a fault response for the provided ID. This will only succeed if another response with the same Fault ID is not already enabled.
- **`Fault_Correction_Instance.Disable_Fault_Response`** - Disable a fault response for the provided ID.
- **`Fault_Correction_Instance.Clear_Fault_Response`** - Resets a fault response to the Enabled state of the provided ID. If the fault is latched, it unlatches the fault.

- **`Fault_Correction_Instance.Clear_All_Fault_Responses`** - Resets all fault responses to the Enabled state. Unlatches all latched fault responses.
- **`Fault_Correction_Instance.Reset_Data_Products`** - This command resets the values of all the component's data product to the values at initialization, except for the Fault_Response_Statuses data product which can be reset by the Clear_All_Fault_Responses command.
- **`Cpu_Monitor_Instance.Set_Packet_Period`** - Set the period of the packet. A period of zero disables the sending of the packet.
- **`Queue_Monitor_Instance.Set_Packet_Period`** - Set the period of the packet. A period of zero disables the sending of the packet.
- **`Stack_Monitor_Instance.Set_Packet_Period`** - Set the period of the packet. A period of zero disables the sending of the packet.
- **`Counter_Instance.Set_Count`** - Change the current counter value in the counter component
- **`Counter_Instance.Reset_Count`** - Reset the current counter value in the counter component to zero
- **`Counter_Instance.Set_Count_Add`** - Change the current counter value in the counter component to the sum of the arguments

2.6 Parameters

The table below shows the parameters for the Pico Example assembly.

Table 5: Pico Example Parameters

Parameter ID	Parameter Name	Type	Default Value
0x0001 (1)	Oscillator_A. Frequency	Packed_F32.T	(Value=>0.175)
0x0002 (2)	Oscillator_A. Amplitude	Packed_F32.T	(Value=>5.0)
0x0003 (3)	Oscillator_A. Offset	Packed_F32.T	(Value=>0.0)
0x0004 (4)	Oscillator_B. Frequency	Packed_F32.T	(Value=>0.175)
0x0005 (5)	Oscillator_B. Amplitude	Packed_F32.T	(Value=>5.0)
0x0006 (6)	Oscillator_B. Offset	Packed_F32.T	(Value=>0.0)

Parameter Descriptions:

- **`Oscillator_A.Frequency`** - The frequency of the oscillator in Hz
- **`Oscillator_A.Amplitude`** - The amplitude of the oscillator
- **`Oscillator_A.Offset`** - The Y offset of the oscillator

- **`Oscillator_B.Frequency`** - The frequency of the oscillator in Hz
- **`Oscillator_B.Amplitude`** - The amplitude of the oscillator
- **`Oscillator_B.Offset`** - The Y offset of the oscillator

2.7 Events

The table below shows the events for the Pico Example assembly.

Table 6: Pico Example Events

Event ID	Event Name	Parameter Type
0x0001 (1)	Tick_Divider_Instance. Component_Has_Full_Queue	Td_Full_Queue_Param.T
0x0002 (2)	Slow_Rate_Group.Cycle_Slip	Cycle_Slip_Param.T
0x0003 (3)	Slow_Rate_Group.Max_Cycle_ Time_Exceeded	Time_Exceeded.T
0x0004 (4)	Slow_Rate_Group.Max_ Execution_Time_Exceeded	Time_Exceeded.T
0x0005 (5)	Slow_Rate_Group.Component_ Has_Full_Queue	Full_Queue_Param.T
0x0006 (6)	Slow_Rate_Group.Incoming_ Tick_Dropped	Tick.T
0x0007 (7)	Fast_Rate_Group.Cycle_Slip	Cycle_Slip_Param.T
0x0008 (8)	Fast_Rate_Group.Max_Cycle_ Time_Exceeded	Time_Exceeded.T
0x0009 (9)	Fast_Rate_Group.Max_ Execution_Time_Exceeded	Time_Exceeded.T
0x000a (10)	Fast_Rate_Group.Component_ Has_Full_Queue	Full_Queue_Param.T
0x000b (11)	Fast_Rate_Group.Incoming_ Tick_Dropped	Tick.T
0x000c (12)	Watchdog_Rate_Group.Cycle_ Slip	Cycle_Slip_Param.T
0x000d (13)	Watchdog_Rate_Group.Max_ Cycle_Time_Exceeded	Time_Exceeded.T

0x000e (14)	Watchdog_Rate_Group.Max_ Execution_Time_Exceeded	Time_Exceeded.T
0x000f (15)	Watchdog_Rate_Group. Component_Has_Full_Queue	Full_Queue_Param.T
0x0010 (16)	Watchdog_Rate_Group. Incoming_Tick_Dropped	Tick.T
0x0011 (17)	Ccsds_Command_Depacketizer_ Instance.Invalid_Packet_ Checksum	Invalid_Packet_Xor8_Info.T
0x0012 (18)	Ccsds_Command_Depacketizer_ Instance.Invalid_Packet_ Type	Ccsds_Primary_Header.T
0x0013 (19)	Ccsds_Command_Depacketizer_ Instance.Packet_Too_Small	Invalid_Packet_Length.T
0x0014 (20)	Ccsds_Command_Depacketizer_ Instance.Packet_Too_Large	Invalid_Packet_Length.T
0x0015 (21)	Ccsds_Command_Depacketizer_ Instance.No_Secondary_ Header	Ccsds_Primary_Header.T
0x0016 (22)	Ccsds_Command_Depacketizer_ Instance.Counts_Reset	-
0x0017 (23)	Ccsds_Command_Depacketizer_ Instance.Invalid_Command_ Received	Invalid_Command_Info.T
0x0018 (24)	Command_Router_Instance. Command_Received	Command_Header.T
0x0019 (25)	Command_Router_Instance. Command_Execution_ Successful	Command_Response.T
0x001a (26)	Command_Router_Instance. Command_Execution_Failure	Command_Response.T
0x001b (27)	Command_Router_Instance. Command_Id_Not_Registered	Command_Header.T
0x001c (28)	Command_Router_Instance. Registration_Id_Conflict	Command_Id.T

0x001d (29)	Command_Router_Instance. Router_Table_Full	Command_Id.T
0x001e (30)	Command_Router_Instance. Outgoing_Command_Dropped	Command_Header.T
0x001f (31)	Command_Router_Instance. Incoming_Command_Dropped	Command_Header.T
0x0020 (32)	Command_Router_Instance. Noop_Command_Dropped	Command_Header.T
0x0021 (33)	Command_Router_Instance. Command_Response_Dropped	Command_Response.T
0x0022 (34)	Command_Router_Instance. Noop_Received	-
0x0023 (35)	Command_Router_Instance. Noop_Arg_Received	Command_Router_Arg.T
0x0024 (36)	Command_Router_Instance. Noop_Response_Received	-
0x0025 (37)	Command_Router_Instance. Noop_Response_Forwarding_ Success	Command_Response.T
0x0026 (38)	Command_Router_Instance. Forwarded_Command_Response_ Dropped	Command_Response.T
0x0027 (39)	Command_Router_Instance. Invalid_Command_Source_Id	Command_Response.T
0x0028 (40)	Command_Router_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x0029 (41)	Command_Router_Instance. Data_Products_Reset	-
0x002a (42)	Event_Filter_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x002b (43)	Event_Filter_Instance. Filtered_Event	Event_Filter_Single_Event_ Cmd_Type.T
0x002c (44)	Event_Filter_Instance. Unfiltered_Event	Event_Filter_Single_Event_ Cmd_Type.T

0x002d (45)	Event_Filter_Instance. Filtered_Event_Range	Event_Filter_Id_Range.T
0x002e (46)	Event_Filter_Instance. Unfiltered_Event_Range	Event_Filter_Id_Range.T
0x002f (47)	Event_Filter_Instance. Enable_Event_Filter	-
0x0030 (48)	Event_Filter_Instance. Disable_Event_Filter	-
0x0031 (49)	Event_Filter_Instance. Filter_Event_Invalid_Id	Event_Filter_Single_Event_ Cmd_Type.T
0x0032 (50)	Event_Filter_Instance. Unfilter_Event_Invalid_Id	Event_Filter_Single_Event_ Cmd_Type.T
0x0033 (51)	Event_Filter_Instance. Filter_Event_Range_Invalid_ Id	Event_Filter_Id_Range.T
0x0034 (52)	Event_Filter_Instance. Unfilter_Event_Range_ Invalid_Id	Event_Filter_Id_Range.T
0x0035 (53)	Event_Filter_Instance.Dump_ Event_States_Received	-
0x0036 (54)	Event_Limiter_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x0037 (55)	Event_Limiter_Instance. Events_Limited_Since_Last_ Tick	Event_Limiter_Num_Events_ Type.T
0x0038 (56)	Event_Limiter_Instance. Event_Limit_Enabled	Event_Single_State_Cmd_ Type.T
0x0039 (57)	Event_Limiter_Instance. Event_Limit_Disabled	Event_Single_State_Cmd_ Type.T
0x003a (58)	Event_Limiter_Instance. Event_Limit_Range_Enabled	Event_Limiter_Id_Range.T
0x003b (59)	Event_Limiter_Instance. Event_Limit_Range_Disabled	Event_Limiter_Id_Range.T
0x003c (60)	Event_Limiter_Instance. Event_Limiting_Enabled	-

0x003d (61)	Event_Limiter_Instance. Event_Limiting_Disabled	-
0x003e (62)	Event_Limiter_Instance. Event_Limit_Enable_Invalid_ Id	Event_Single_State_Cmd_ Type.T
0x003f (63)	Event_Limiter_Instance. Event_Limit_Disable_ Invalid_Id	Event_Single_State_Cmd_ Type.T
0x0040 (64)	Event_Limiter_Instance. Event_Limit_Range_Enabled_ Invalid_Id	Event_Limiter_Id_Range.T
0x0041 (65)	Event_Limiter_Instance. Event_Limit_Range_Disabled_ Invalid_Id	Event_Limiter_Id_Range.T
0x0042 (66)	Event_Limiter_Instance.Set_ New_Persistence	Event_Limiter_Persistence_ Type.T
0x0043 (67)	Event_Limiter_Instance. Dump_Event_States_Received	-
0x0044 (68)	Product_Database_Instance. Data_Product_Update_Id_Out_ Of_Range	Data_Product_Id.T
0x0045 (69)	Product_Database_Instance. Data_Product_Fetch_Id_Out_ Of_Range	Data_Product_Id.T
0x0046 (70)	Product_Database_Instance. Data_Product_Fetch_Id_Not_ Available	Data_Product_Id.T
0x0047 (71)	Product_Database_Instance. Override_Cleared	Data_Product_Id.T
0x0048 (72)	Product_Database_Instance. Override_Cleared_For_All	-
0x0049 (73)	Product_Database_Instance. Data_Product_Overridden	Data_Product_Header.T
0x004a (74)	Product_Database_Instance. Data_Product_Override_ Serialization_Failure	Data_Product_Header.T

0x004b (75)	Product_Database_Instance. Data_Product_Override_Id_ Out_Of_Range	Data_Product_Id.T
0x004c (76)	Product_Database_Instance. Data_Product_Clear_ Override_Id_Out_Of_Range	Data_Product_Id.T
0x004d (77)	Product_Database_Instance. Data_Product_Dump_Id_Not_ Available	Data_Product_Id.T
0x004e (78)	Product_Database_Instance. Data_Product_Dump_Id_Out_ Of_Range	Data_Product_Id.T
0x004f (79)	Product_Database_Instance. Data_Product_Dumped	Data_Product_Header.T
0x0050 (80)	Product_Database_Instance. Dumping_Data_Product_Poly_ Type	Data_Product_Poly_Extract.T
0x0051 (81)	Product_Database_Instance. Dumped_Data_Product_Poly_ Type	Data_Product_Poly_Event.T
0x0052 (82)	Product_Database_Instance. Data_Product_Dump_Poly_Id_ Not_Available	Data_Product_Id.T
0x0053 (83)	Product_Database_Instance. Data_Product_Dump_Poly_Id_ Out_Of_Range	Data_Product_Id.T
0x0054 (84)	Product_Database_Instance. Data_Product_Poly_Type_ Extraction_Failed	Data_Product_Header.T
0x0055 (85)	Product_Database_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x0056 (86)	Product_Packetizer_ Instance.Invalid_Packet_ Id_Commanded	Invalid_Packet_Id.T
0x0057 (87)	Product_Packetizer_ Instance.Packet_Enabled	Packet_Period.T
0x0058 (88)	Product_Packetizer_ Instance.Packet_Disabled	Packet_Period.T

0x0059 (89)	Product_Packetizer_ Instance.Packet_Enabled_ On_Change	Packet_Period.T
0x005a (90)	Product_Packetizer_ Instance.Packet_Period_Set	Packet_Period.T
0x005b (91)	Product_Packetizer_ Instance.Data_Product_ Missing_On_Fetch	Packet_Data_Product_Ids.T
0x005c (92)	Product_Packetizer_ Instance.Packet_Period_ Item_Bad_Id	Packet_Data_Product_Ids.T
0x005d (93)	Product_Packetizer_ Instance.Data_Product_ Length_Mismatch	Invalid_Data_Product_ Length.T
0x005e (94)	Product_Packetizer_ Instance.Invalid_Command_ Received	Invalid_Command_Info.T
0x005f (95)	Product_Packetizer_ Instance.Dropped_Command	Command_Header.T
0x0060 (96)	Ccsds_Serial_Interface_ Instance.Packet_Send_Failed	Ccsds_Primary_Header.T
0x0061 (97)	Ccsds_Serial_Interface_ Instance.Packet_Recv_Failed	Ccsds_Primary_Header.T
0x0062 (98)	Ccsds_Serial_Interface_ Instance.Have_Not_Seen_ Sync_Pattern	Packed_U32.T
0x0063 (99)	Counter_Instance.Set_Count_ Command_Received	Packed_U32.T
0x0064 (100)	Counter_Instance.Reset_ Count_Command_Received	-
0x0065 (101)	Counter_Instance.Set_Count_ Add_Command_Received	Operands.T
0x0066 (102)	Counter_Instance.Sending_ Value	Packed_U32.T
0x0067 (103)	Counter_Instance.Dropped_ Command	Command_Header.T

0x0068 (104)	Counter_Instance.Invalid_Command_Received	Invalid_Command_Info.T
0x0069 (105)	Oscillator_A.Frequency_Value_Set	Packed_F32.T
0x006a (106)	Oscillator_A.Amplitude_Value_Set	Packed_F32.T
0x006b (107)	Oscillator_A.Offset_Value_Set	Packed_F32.T
0x006c (108)	Oscillator_A.Dropped_Command	Command_Header.T
0x006d (109)	Oscillator_A.Invalid_Command_Received	Invalid_Command_Info.T
0x006e (110)	Oscillator_A.Invalid_Parameter_Received	Invalid_Parameter_Info.T
0x006f (111)	Oscillator_B.Frequency_Value_Set	Packed_F32.T
0x0070 (112)	Oscillator_B.Amplitude_Value_Set	Packed_F32.T
0x0071 (113)	Oscillator_B.Offset_Value_Set	Packed_F32.T
0x0072 (114)	Oscillator_B.Dropped_Command	Command_Header.T
0x0073 (115)	Oscillator_B.Invalid_Command_Received	Invalid_Command_Info.T
0x0074 (116)	Oscillator_B.Invalid_Parameter_Received	Invalid_Parameter_Info.T
0x0075 (117)	Zero_Divider_Instance.Dividing_By_Zero	Packed_Natural.T
0x0076 (118)	Zero_Divider_Instance.Invalid_Magic_Number	Packed_U32.T
0x0077 (119)	Zero_Divider_Instance.Invalid_Command_Received	Invalid_Command_Info.T
0x0078 (120)	Task_Watchdog_Instance.Watchdog_Pet_Checks_Enabled	-

0x0079 (121)	Task_Watchdog_Instance. Watchdog_Pet_Checks_ Disabled	-
0x007a (122)	Task_Watchdog_Instance. Watchdog_Limit_Set	Watchdog_Limit_Cmd.T
0x007b (123)	Task_Watchdog_Instance. Watchdog_Action_Set	Watchdog_Action_Cmd.T
0x007c (124)	Task_Watchdog_Instance. Watchdog_Limit_Change_ Index_Out_Of_Range	Packed_Connector_Index.T
0x007d (125)	Task_Watchdog_Instance. Watchdog_Action_Change_ Index_Out_Of_Range	Packed_Connector_Index.T
0x007e (126)	Task_Watchdog_Instance. Watchdog_Action_Change_ Invalid_Transition_To_Fault	Packed_Connector_Index.T
0x007f (127)	Task_Watchdog_Instance. Component_Exceeded_Pet_ Limit	Packed_Connector_Index.T
0x0080 (128)	Task_Watchdog_Instance. Critical_Task_Not_Petting	Packed_Connector_Index.T
0x0081 (129)	Task_Watchdog_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x0082 (130)	Fault_Producer_Instance. Sending_Fault_1	-
0x0083 (131)	Fault_Producer_Instance. Sending_Fault_2	-
0x0084 (132)	Fault_Producer_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x0085 (133)	Fault_Correction_Instance. Fault_Received	Fault_Static.T
0x0086 (134)	Fault_Correction_Instance. Fault_Response_Sent	Command_Header.T
0x0087 (135)	Fault_Correction_Instance. Fault_Response_Cleared	Packed_Fault_Id.T

0x0088 (136)	Fault_Correction_Instance. Fault_Response_Disabled	Packed_Fault_Id.T
0x0089 (137)	Fault_Correction_Instance. Fault_Response_Enabled	Packed_Fault_Id.T
0x008a (138)	Fault_Correction_Instance. All_Fault_Responses_Cleared	-
0x008b (139)	Fault_Correction_Instance. Unrecognized_Fault_Id	Packed_Fault_Id.T
0x008c (140)	Fault_Correction_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x008d (141)	Fault_Correction_Instance. Command_Dropped	Command_Header.T
0x008e (142)	Fault_Correction_Instance. Fault_Dropped	Fault_Header.T
0x008f (143)	Fault_Correction_Instance. Data_Products_Reset	-
0x0090 (144)	Cpu_Monitor_Instance. Packet_Period_Set	Packed_U16.T
0x0091 (145)	Cpu_Monitor_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x0092 (146)	Queue_Monitor_Instance. Packet_Period_Set	Packed_U16.T
0x0093 (147)	Queue_Monitor_Instance. Invalid_Command_Received	Invalid_Command_Info.T
0x0094 (148)	Stack_Monitor_Instance. Packet_Period_Set	Packed_U16.T
0x0095 (149)	Stack_Monitor_Instance. Invalid_Command_Received	Invalid_Command_Info.T

Event Descriptions:

- **`Tick_Divider_Instance.Component_Has_Full_Queue`** - The tick divider tried to put a Tick on a component's queue, but the queue was full, so the Tick was dropped.
- **`Slow_Rate_Group.Cycle_Slip`** - Execution ran long on this cycle.
- **`Slow_Rate_Group.Max_Cycle_Time_Exceeded`** - A new maximum cycle time was reached. The event parameter is a Tick type with the maximum cycle time as the Time and the cycle count where the maximum cycle was achieved as the Count.
- **`Slow_Rate_Group.Max_Execution_Time_Exceeded`** - A new maximum execution time

was reached. The event parameter is a Tick type with the maximum cycle time as the Time and the cycle count where the maximum cycle was achieved as the Count.

- **`Slow_Rate_Group.Component_Has_Full_Queue`** - The rate group tried to put a Tick on a component's queue, but the queue was full, so the Tick was dropped.
- **`Slow_Rate_Group.Incoming_Tick_Dropped`** - The rate group component's queue is full, so it cannot store the tick coming in. This usually means the rate group is cycle slipping and not running as fast as it needs to.
- **`Fast_Rate_Group.Cycle_Slip`** - Execution ran long on this cycle.
- **`Fast_Rate_Group.Max_Cycle_Time_Exceeded`** - A new maximum cycle time was reached. The event parameter is a Tick type with the maximum cycle time as the Time and the cycle count where the maximum cycle was achieved as the Count.
- **`Fast_Rate_Group.Max_Execution_Time_Exceeded`** - A new maximum execution time was reached. The event parameter is a Tick type with the maximum cycle time as the Time and the cycle count where the maximum cycle was achieved as the Count.
- **`Fast_Rate_Group.Component_Has_Full_Queue`** - The rate group tried to put a Tick on a component's queue, but the queue was full, so the Tick was dropped.
- **`Fast_Rate_Group.Incoming_Tick_Dropped`** - The rate group component's queue is full, so it cannot store the tick coming in. This usually means the rate group is cycle slipping and not running as fast as it needs to.
- **`Watchdog_Rate_Group.Cycle_Slip`** - Execution ran long on this cycle.
- **`Watchdog_Rate_Group.Max_Cycle_Time_Exceeded`** - A new maximum cycle time was reached. The event parameter is a Tick type with the maximum cycle time as the Time and the cycle count where the maximum cycle was achieved as the Count.
- **`Watchdog_Rate_Group.Max_Execution_Time_Exceeded`** - A new maximum execution time was reached. The event parameter is a Tick type with the maximum cycle time as the Time and the cycle count where the maximum cycle was achieved as the Count.
- **`Watchdog_Rate_Group.Component_Has_Full_Queue`** - The rate group tried to put a Tick on a component's queue, but the queue was full, so the Tick was dropped.
- **`Watchdog_Rate_Group.Incoming_Tick_Dropped`** - The rate group component's queue is full, so it cannot store the tick coming in. This usually means the rate group is cycle slipping and not running as fast as it needs to.
- **`Ccsds_Command_Depacketizer_Instance.Invalid_Packet_Checksum`** - A packet was received with an invalid checksum
- **`Ccsds_Command_Depacketizer_Instance.Invalid_Packet_Type`** - A packet was received with an invalid ccsds packet type. The expected packet type is a telecommand, but a telemetry packet was received.
- **`Ccsds_Command_Depacketizer_Instance.Packet_Too_Small`** - The packet received was too small to contain necessary command information.
- **`Ccsds_Command_Depacketizer_Instance.Packet_Too_Large`** - The packet received was too large and is bigger than the size of a command.
- **`Ccsds_Command_Depacketizer_Instance.No_Secondary_Header`** - A packet was received without a secondary header, but the secondary header is required.
- **`Ccsds_Command_Depacketizer_Instance.Counts_Reset`** - A command was received to reset the counts.
- **`Ccsds_Command_Depacketizer_Instance.Invalid_Command_Received`** - A command was received with invalid parameters.
- **`Command_Router_Instance.Command_Received`** - A command was received by the command router to be routed.

- **Command_Router_Instance.Command_Execution_Successful** - A command was routed, executed, and returned a response saying it was executed successfully
- **Command_Router_Instance.Command_Execution_Failure** - A command execution failed.
- **Command_Router_Instance.Command_Id_Not_Registered** - A command was sent to the router, but it was not found in the router table.
- **Command_Router_Instance.Registration_Id_Conflict** - The command Id has already been registered.
- **Command_Router_Instance.Router_Table_Full** - Cannot add command Id to router table because it is full.
- **Command_Router_Instance.Outgoing_Command_Dropped** - A command was dropped because the recipient's queue was full.
- **Command_Router_Instance.Incoming_Command_Dropped** - A command was dropped because the command router's queue was full.
- **Command_Router_Instance.Noop_Command_Dropped** - A noop command was dropped because the command router's queue was full.
- **Command_Router_Instance.Command_Response_Dropped** - A command response was dropped because the command router's queue was full.
- **Command_Router_Instance.Noop_Received** - A Noop command was received.
- **Command_Router_Instance.Noop_Arg_Received** - A Noop command was received with an argument.
- **Command_Router_Instance.Noop_Response_Received** - A noop response self test command was received.
- **Command_Router_Instance.Noop_Response_Forwarding_Success** - If this event is sent then the noop response self test command succeeded.
- **Command_Router_Instance.Forwarded_Command_Response_Dropped** - A forwarded command response was dropped because the receiving component's queue overflowed.
- **Command_Router_Instance.Invalid_Command_Source_Id** - A command response contained an invalid source id. This is a software bug and should be corrected.
- **Command_Router_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Command_Router_Instance.Data_Products_Reset** - The component's data products have been reset to initialization values.
- **Event_Filter_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Event_Filter_Instance.Filtered_Event** - This event indicates that the state of an event was set to enabled for the filter.
- **Event_Filter_Instance.Unfiltered_Event** - This event indicates that the state of an event was set to disabled for the filter.
- **Event_Filter_Instance.Filtered_Event_Range** - This event indicates that the state of a range of events were set to enabled for the filter.
- **Event_Filter_Instance.Unfiltered_Event_Range** - This event indicates that the state of a range of events were set to disabled for the filter.
- **Event_Filter_Instance.Enable_Event_Filter** - This event indicates that the state of all events were set to enabled for the filter, but kept the internal state.
- **Event_Filter_Instance.Disable_Event_Filter** - This event indicates that the state of all events were set to disabled for the filter, but kept the internal state.

- **`Event_Filter_Instance.Filter_Event_Invalid_Id`** - This event indicates that the command to change the event state to enabled failed since the event ID was out of range.
- **`Event_Filter_Instance.Unfilter_Event_Invalid_Id`** - This event indicates that the command to change the event state to disabled failed since the event ID was out of range.
- **`Event_Filter_Instance.Filter_Event_Range_Invalid_Id`** - This event indicates that changing the state for the range to enabled, failed due to an invalid id.
- **`Event_Filter_Instance.Unfilter_Event_Range_Invalid_Id`** - This event indicates that changing the state for the range to disabled, failed due to an invalid id.
- **`Event_Filter_Instance.Dump_Event_States_Received`** - Event that indicates the process of building the packet that stores the event states has started and will send the packet once we go through a decrement cycle.
- **`Event_Limiter_Instance.Invalid_Command_Received`** - A command was received with invalid parameters.
- **`Event_Limiter_Instance.Events_Limited_Since_Last_Tick`** - An event that indicates how many events have been limited as well as up to the first 10 ids of those events. The event ids listed may not have been dropped, but are at listed since they are at the current max limit.
- **`Event_Limiter_Instance.Event_Limit_Enabled`** - This event indicates that the state of an event was set to enabled for the limiter.
- **`Event_Limiter_Instance.Event_Limit_Disabled`** - This event indicates that the state of an event was set to disabled for the limiter.
- **`Event_Limiter_Instance.Event_Limit_Range_Enabled`** - This event indicates that the state of a range of events were set to enabled for the limiter.
- **`Event_Limiter_Instance.Event_Limit_Range_Disabled`** - This event indicates that the state of a range of events were set to disabled for the limiter.
- **`Event_Limiter_Instance.Event_Limiting_Enabled`** - This event indicates that the state of all events were set to enabled for the limiter.
- **`Event_Limiter_Instance.Event_Limiting_Disabled`** - This event indicates that the state of all events were set to disabled for the limiter.
- **`Event_Limiter_Instance.Event_Limit_Enable_Invalid_Id`** - This event indicates that the command to change the event state to enabled failed since the event ID was out of range.
- **`Event_Limiter_Instance.Event_Limit_Disable_Invalid_Id`** - This event indicates that the command to change the event state to disabled failed since the event ID was out of range.
- **`Event_Limiter_Instance.Event_Limit_Range_Enabled_Invalid_Id`** - This event indicates that changing the state for the range to enabled, failed due to an invalid id.
- **`Event_Limiter_Instance.Event_Limit_Range_Disabled_Invalid_Id`** - This event indicates that changing the state for the range to disabled, failed due to an invalid id.
- **`Event_Limiter_Instance.Set_New_Persistence`** - Indicates that the persistence of the number of events until we limit was changed to a new value between 1 and 7.
- **`Event_Limiter_Instance.Dump_Event_States_Received`** - Event that indicates the process of building the packet that stores the event states has started and will send the packet once we go through a decrement cycle.
- **`Product_Database_Instance.Data_Product_Update_Id_Out_Of_Range`** - A data product update was received with an ID that was out of range.
- **`Product_Database_Instance.Data_Product_Fetch_Id_Out_Of_Range`** - A data product fetch was received with an ID that was out of range.

- **Product_Database_Instance.Data_Product_Fetch_Id_Not_Available** - A data product fetch was received with an ID that has not yet been stored in the database.
- **Product_Database_Instance.Override_Cleared** - Override condition cleared for the data product of the provided ID.
- **Product_Database_Instance.Override_Cleared_For_All** - Override condition cleared for all data products.
- **Product_Database_Instance.Data_Product_Overridden** - Data product overridden by command.
- **Product_Database_Instance.Data_Product_Override_Serialization_Failure** - Data product override could not be completed due to a serialization error.
- **Product_Database_Instance.Data_Product_Override_Id_Out_Of_Range** - A data product override command was received with an ID that was out of range.
- **Product_Database_Instance.Data_Product_Clear_Override_Id_Out_Of_Range** - A data product clear override command was received with an ID that was out of range.
- **Product_Database_Instance.Data_Product_Dump_Id_Not_Available** - A data product dump command was received with an ID that has not yet been stored in the database.
- **Product_Database_Instance.Data_Product_Dump_Id_Out_Of_Range** - A data product dump command was received with an ID that was out of range.
- **Product_Database_Instance.Data_Product_Dumped** - Data product dumped into a packet by command.
- **Product_Database_Instance.Dumping_Data_Product_Poly_Type** - Data product poly type dumped into a packet by command.
- **Product_Database_Instance.Dumped_Data_Product_Poly_Type** - Data product poly type dumped into a packet by command.
- **Product_Database_Instance.Data_Product_Dump_Poly_Id_Not_Available** - A data product dump poly command was received with an ID that has not yet been stored in the database.
- **Product_Database_Instance.Data_Product_Dump_Poly_Id_Out_Of_Range** - A data product dump poly command was received with an ID that was out of range.
- **Product_Database_Instance.Data_Product_Poly_Type_Extraction_Failed** - A data product dump poly command failed because the extraction could not succeed with the provided parameters.
- **Product_Database_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Product_Packetizer_Instance.Invalid_Packet_Id_Commanded** - An invalid packet id was commanded for a given command.
- **Product_Packetizer_Instance.Packet_Enabled** - A packet was enabled.
- **Product_Packetizer_Instance.Packet_Disabled** - A packet was disabled.
- **Product_Packetizer_Instance.Packet_Enabled_On_Change** - A packet was enabled in on-change mode.
- **Product_Packetizer_Instance.Packet_Period_Set** - A packet period was set.
- **Product_Packetizer_Instance.Data_Product_Missing_On_Fetch** - A data product was missing when fetched for packet insertion.
- **Product_Packetizer_Instance.Packet_Period_Item_Bad_Id** - A packet period packet item could not be formed because the ID is invalid.
- **Product_Packetizer_Instance.Data_Product_Length_Mismatch** - A data product

was fetched but contained an unexpected length.

- **Product_Packetizer_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Product_Packetizer_Instance.Dropped_Command** - A command was dropped due to a full queue.
- **Ccsds_Serial_Interface_Instance.Packet_Send_Failed** - Failed to send a packet over the serial port because it has an invalid CCSDS header.
- **Ccsds_Serial_Interface_Instance.Packet_Recv_Failed** - Failed to receive a packet over the serial port because it has an invalid CCSDS header.
- **Ccsds_Serial_Interface_Instance.Have_Not_Seen_Sync_Pattern** - The component has received N number of bytes without seeing a sync pattern yet.
- **Counter_Instance.Set_Count_Command_Received** - Received a Set_Count command.
- **Counter_Instance.Reset_Count_Command_Received** - Received a Reset_Count command.
- **Counter_Instance.Set_Count_Add_Command_Received** - Received a Set_Count_Add command.
- **Counter_Instance.Sending_Value** - Sending the current value out as data product.
- **Counter_Instance.Dropped_Command** - The component's queue overflowed and the command was dropped.
- **Counter_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Oscillator_A.Frequency_Value_Set** - A new frequency value was set by command
- **Oscillator_A.Amplitude_Value_Set** - A new amplitude value was set by command
- **Oscillator_A.Offset_Value_Set** - A new offset value was set by command
- **Oscillator_A.Dropped_Command** - The component's queue overflowed and the command was dropped.
- **Oscillator_A.Invalid_Command_Received** - A command was received with invalid parameters.
- **Oscillator_A.Invalid_Parameter_Received** - A parameter was received with invalid parameters.
- **Oscillator_B.Frequency_Value_Set** - A new frequency value was set by command
- **Oscillator_B.Amplitude_Value_Set** - A new amplitude value was set by command
- **Oscillator_B.Offset_Value_Set** - A new offset value was set by command
- **Oscillator_B.Dropped_Command** - The component's queue overflowed and the command was dropped.
- **Oscillator_B.Invalid_Command_Received** - A command was received with invalid parameters.
- **Oscillator_B.Invalid_Parameter_Received** - A parameter was received with invalid parameters.
- **Zero_Divider_Instance.Dividing_By_Zero** - A divide by zero command was received, and the magic number was correct. The division will occur in N milliseconds, where N is provided as the event parameter.
- **Zero_Divider_Instance.Invalid_Magic_Number** - A divide by zero command was received, but the magic number was incorrect. The division will not occur.
- **Zero_Divider_Instance.Invalid_Command_Received** - A command was received with invalid parameters.

- **Task_Watchdog_Instance.Watchdog_Pet_Checks_Enabled** - Indicates a command was received to enable the checks on upstream pets.
- **Task_Watchdog_Instance.Watchdog_Pet_Checks_Disabled** - Indicates a command was received to disable the checks on upstream pets.
- **Task_Watchdog_Instance.Watchdog_Limit_Set** - An event to indicate that the limit was changed by command for a particular index.
- **Task_Watchdog_Instance.Watchdog_Action_Set** - An event to indicate that the action was changed by command for a particular index.
- **Task_Watchdog_Instance.Watchdog_Limit_Change_Index_Out_Of_Range** - Event indicating there was an error for the index range in the set limit command.
- **Task_Watchdog_Instance.Watchdog_Action_Change_Index_Out_Of_Range** - Event indicating there was an error for the index range in the set action command.
- **Task_Watchdog_Instance.Watchdog_Action_Change_Invalid_Transition_To_Fault** - Event indicating there was an error trying to set the action to fault. The petter did not have a fault declared in the model so the action cannot be set to fault.
- **Task_Watchdog_Instance.Component_Exceeded_Pet_Limit** - Event to indicate a pet connector has not received a pet within the set limits for that component.
- **Task_Watchdog_Instance.Critical_Task_Not_Petting** - Event to indicate that one or more of our critical tasks have not indicated a pet in the maximum limit of ticks. The hardware watchdog will not be pet in this case.
- **Task_Watchdog_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Fault_Producer_Instance.Sending_Fault_1** - The component received a command to send out fault 1.
- **Fault_Producer_Instance.Sending_Fault_2** - The component received a command to send out fault 2.
- **Fault_Producer_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Fault_Correction_Instance.Fault_Received** - A fault was received.
- **Fault_Correction_Instance.Fault_Response_Sent** - A fault response was sent with the included command header.
- **Fault_Correction_Instance.Fault_Response_Cleared** - A fault response was cleared.
- **Fault_Correction_Instance.Fault_Response_Disabled** - A fault response has been disabled
- **Fault_Correction_Instance.Fault_Response_Enabled** - A fault response has been enabled.
- **Fault_Correction_Instance.All_Fault_Responses_Cleared** - Any latched faults have been unlatched by command.
- **Fault_Correction_Instance.Unrecognized_Fault_Id** - A fault response entry with the included fault ID was not found in the table.
- **Fault_Correction_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Fault_Correction_Instance.Command_Dropped** - A command was dropped due to a full queue.
- **Fault_Correction_Instance.Fault_Dropped** - A fault was dropped due to a full queue.
- **Fault_Correction_Instance.Data_Products_Reset** - The component's data prod-

ucts have been reset to initialization values.

- **Cpu_Monitor_Instance.Packet_Period_Set** - A command was received to change the packet period.
- **Cpu_Monitor_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Queue_Monitor_Instance.Packet_Period_Set** - A command was received to change the packet period.
- **Queue_Monitor_Instance.Invalid_Command_Received** - A command was received with invalid parameters.
- **Stack_Monitor_Instance.Packet_Period_Set** - A command was received to change the packet period.
- **Stack_Monitor_Instance.Invalid_Command_Received** - A command was received with invalid parameters.

2.8 Data Products

The table below shows the data products for the Pico Example assembly.

Table 7: Pico Example Data Products

Data Product ID	Data Product Name	Type
0x0001 (1)	Slow_Rate_Group.Timing_Report	Task_Timing_Report.T
0x0002 (2)	Fast_Rate_Group.Timing_Report	Task_Timing_Report.T
0x0003 (3)	Watchdog_Rate_Group.Timing_Report	Task_Timing_Report.T
0x0004 (4)	Ccsds_Command_Depacketizer_Instance.Rejected_Packet_Count	Packed_U16.T
0x0005 (5)	Ccsds_Command_Depacketizer_Instance.Accepted_Packet_Count	Packed_U16.T
0x0006 (6)	Command_Router_Instance.Command_Receive_Count	Packed_U16.T
0x0007 (7)	Command_Router_Instance.Command_Success_Count	Packed_U16.T
0x0008 (8)	Command_Router_Instance.Command_Failure_Count	Packed_U16.T
0x0009 (9)	Command_Router_Instance.Last_Received_Command	Command_Id.T

0x000a (10)	Command_Router_Instance. Last_Successful_Command	Command_Id.T
0x000b (11)	Command_Router_Instance. Last_Failed_Command	Command_Id_Status.T
0x000c (12)	Command_Router_Instance. Noop_Arg_Last_Value	Command_Router_Arg.T
0x000d (13)	Event_Filter_Instance. Total_Events_Filtered	Packed_U32.T
0x000e (14)	Event_Filter_Instance. Total_Events_Unfiltered	Packed_U32.T
0x000f (15)	Event_Filter_Instance. Component_Filter_State	Event_Component_State_ Type.T
0x0010 (16)	Event_Limiter_Instance. Limited_Events_Since_Tick	Packed_U16.T
0x0011 (17)	Event_Limiter_Instance. Total_Events_Limited	Packed_U32.T
0x0012 (18)	Event_Limiter_Instance. Component_Limiting_ Enabled_Status	Event_Enable_State_Type.T
0x0013 (19)	Event_Packetizer_Instance. Events_Dropped_Count	Packed_U32.T
0x0014 (20)	Event_Packetizer_Instance. Bytes_Available	Packed_Natural.T
0x0015 (21)	Product_Database_Instance. Data_Product_Poly_Type_ Dump	Data_Product_Poly_Type.T
0x0016 (22)	Product_Database_Instance. Database_Override	Packed_Enable_Disable_ Type.T
0x0017 (23)	Oscillator_A.Oscillator_ Value	Packed_F32.T
0x0018 (24)	Oscillator_B.Oscillator_ Value	Packed_F32.T
0x0019 (25)	Adc_Data_Collector_ Instance.Channel_0	Packed_Integer.T

0x001a (26)	Adc_Data_Collector_ Instance.Vsys	Packed_Integer.T
0x001b (27)	Adc_Data_Collector_ Instance.Temperature	Packed_Integer.T
0x001c (28)	Task_Watchdog_Instance. Watchdog_Component_Petter_ State	Packed_Watchdog_Component_ State.T
0x001d (29)	Task_Watchdog_Instance. Pet_Connector_Action_ States	Pico_Example_Task_ Watchdog_List_State_ Record.T
0x001e (30)	Task_Watchdog_Instance. Slow_Rate_Group_Limit	Packed_Missed_Pet_Limit.T
0x001f (31)	Task_Watchdog_Instance. Fast_Rate_Group_Limit	Packed_Missed_Pet_Limit.T
0x0020 (32)	Fault_Correction_Instance. Fault_Counter	Packed_U16.T
0x0021 (33)	Fault_Correction_Instance. Last_Fault_Id_Received	Pico_Example_Fault_ Responses_Packed_Id_Type.T
0x0022 (34)	Fault_Correction_Instance. Time_Of_Last_Fault_ Received	Sys_Time.T
0x0023 (35)	Fault_Correction_Instance. Fault_Response_Statuses	Pico_Example_Fault_ Responses_Status_Record.T
0x0024 (36)	Cpu_Monitor_Instance. Packet_Period	Packed_U16.T
0x0025 (37)	Queue_Monitor_Instance. Packet_Period	Packed_U16.T
0x0026 (38)	Stack_Monitor_Instance. Packet_Period	Packed_U16.T

Data Product Descriptions:

- **Slow_Rate_Group.Timing_Report** - Data relating to timing performance of the component.
- **Fast_Rate_Group.Timing_Report** - Data relating to timing performance of the component.
- **Watchdog_Rate_Group.Timing_Report** - Data relating to timing performance of the component.
- **Ccsds_Command_Depacketizer_Instance.Rejected_Packet_Count** - The number of

packets rejected by the component due to invalid data

- **Ccsds_Command_Depacketizer_Instance.Accepted_Packet_Count** - The number of packets accepted by the component
- **Command_Router_Instance.Command_Receive_Count** - The number of commands received by the component.
- **Command_Router_Instance.Command_Success_Count** - The number of commands that successfully executed.
- **Command_Router_Instance.Command_Failure_Count** - The number of commands that failed to execute.
- **Command_Router_Instance.Last_Received_Command** - The ID of the last received command by the command router.
- **Command_Router_Instance.Last_Successful_Command** - The ID of the last successful command routed by the command router.
- **Command_Router_Instance.Last_Failed_Command** - The ID and status of the last failed command routed by the command router.
- **Command_Router_Instance.Noop_Arg_Last_Value** - The last value sent with the Noop_Arg command. This data product can be useful for testing purposes.
- **Event_Filter_Instance.Total_Events_Filtered** - The total number of events that have been filtered for the components lifetime.
- **Event_Filter_Instance.Total_Events_Unfiltered** - The total number of events that have been passed through for the components lifetime. Does not include out of range IDs.
- **Event_Filter_Instance.Component_Filter_State** - The state of the master switch for filtering events.
- **Event_Limiter_Instance.Limited_Events_Since_Tick** - The number of events that were limited since the last tick.
- **Event_Limiter_Instance.Total_Events_Limited** - The total number of events that have been limited for the components lifetime.
- **Event_Limiter_Instance.Component_Limiting_Enabled_Status** - The current state of the component master switch.
- **Event_Packetizer_Instance.Events_Dropped_Count** - The number of events dropped by the component.
- **Event_Packetizer_Instance.Bytes_Available** - The current number of bytes available for event storage within the component.
- **Product_Database_Instance.Data_Product_Poly_Type_Dump** - Data product poly type dumped into a data product by command.
- **Product_Database_Instance.Database_Override** - If set to Enabled then the database contains at least one data product that has been overridden by command.
- **Oscillator_A.Oscillator_Value** - The current value of the oscillator.
- **Oscillator_B.Oscillator_Value** - The current value of the oscillator.
- **Adc_Data_Collector_Instance.Channel_0** - The ADC reading at Channel 0 in microvolts.
- **Adc_Data_Collector_Instance.Vsys** - The ADC reading of the system voltage in microvolts.
- **Adc_Data_Collector_Instance.Temperature** - The ADC temperature reading in Celsius.
- **Task_Watchdog_Instance.Watchdog_Component_Petter_State** - Data product that tracks the global state to enable or disable all checks on the upstream watchdog pets.

- **Task_Watchdog_Instance.Pet_Connector_Action_States** - 2-bit of state for each pet connector indicating the current action that will be taken if there is an error. Note that Packed_U32.T is just a placeholder type for this data product. The actual type of this data product will be autocoded and at assembly model ingest time.
- **Task_Watchdog_Instance.Slow_Rate_Group_Limit** - Slow rate group monitoring.
- **Task_Watchdog_Instance.Fast_Rate_Group_Limit** - Fast rate group monitoring.
- **Fault_Correction_Instance.Fault_Counter** - The number of faults received by the component.
- **Fault_Correction_Instance.Last_Fault_Id_Received** - The ID of the last fault received.
- **Fault_Correction_Instance.Time_Of_Last_Fault_Received** - The system time of the last fault received.
- **Fault_Correction_Instance.Fault_Response_Statuses** - 2-bits of status for each fault response that this component is managing. Note that Packed_U32.T is just a placeholder type for this data product. The actual type of this data product will be autocoded and at assembly model ingest time.
- **Cpu_Monitor_Instance.Packet_Period** - The current packet period.
- **Queue_Monitor_Instance.Packet_Period** - The current packet period.
- **Stack_Monitor_Instance.Packet_Period** - The current packet period.

2.9 Packets

The table below shows the packets for the Pico Example assembly.

Table 8: Pico Example Packets

Packet ID	Packet Name	Type
0x0001 (1)	Product_Packetizer_Instance.Housekeeping_Packet	*Undefined*
0x0002 (2)	Product_Packetizer_Instance.Software_Status_Packet	*Undefined*
0x0003 (3)	Ccsds_Command_Depacketizer_Instance.Error_Packet	Ccsds_Space_Packet.T
0x0004 (4)	Event_Filter_Instance.Event_Filter_State_Packet	*Undefined*
0x0005 (5)	Event_Limiter_Instance.Event_Limiter_State_Packet	*Undefined*
0x0006 (6)	Product_Database_Instance.Dump_Packet	Data_Product.T

0x0007 (7)	Counter_Instance.Counter_Value	Packed_U32.T
0x0008 (8)	Cpu_Monitor_Instance.Cpu_Usage_Packet	Pico_Example_Cpu_Monitor_Packet_Type.T
0x0009 (9)	Queue_Monitor_Instance.Queue_Usage_Packet	Pico_Example_Queue_Monitor_Packet_Type.T
0x000a (10)	Stack_Monitor_Instance.Stack_Usage_Packet	Pico_Example_Stack_Monitor_Packet_Type.T
0x0061 (97)	Zero_Divider_Instance.Last_Chance_Handler_Packet	Packed_Exception_Occurrence.T
0x0062 (98)	Event_Packetizer_Instance.Events_Packet	*Undefined*

Packet Descriptions:

- **Product_Packetizer_Instance.Housekeeping_Packet** - This packet contains housekeeping data.
- **Product_Packetizer_Instance.Software_Status_Packet** - This packet contains software status information.
- **Ccsds_Command_Depacketizer_Instance.Error_Packet** - This packet contains a CCSDS packet that was dropped due to error.
- **Event_Filter_Instance.Event_Filter_State_Packet** - The packet used to dump all the state information for which events are filtered and which are not. Each event ID takes a bit and any extra bits beyond the event range will show as not filtered.
- **Event_Limiter_Instance.Event_Limiter_State_Packet** - The packet used to dump all the state information for which events are limited and which are not. Each event takes a bit and any extra bits beyond the event range will show as disabled.
- **Product_Database_Instance.Dump_Packet** - This packet contains dumped data products.
- **Counter_Instance.Counter_Value** - The counter value 1.
- **Cpu_Monitor_Instance.Cpu_Usage_Packet** - This packet contains cpu usage numbers for tasks and interrupts in the system.
- **Queue_Monitor_Instance.Queue_Usage_Packet** - This packet contains queue usage numbers for queued components in the system.
- **Stack_Monitor_Instance.Stack_Usage_Packet** - This packet contains stack and secondary stack usage numbers for tasks in the system.
- **Zero_Divider_Instance.Last_Chance_Handler_Packet** - This packet contains information regarding an exception occurrence that triggers the Last_Chance_Handler to get invoked. This packet is not produced directly by this component, and should be produced by the last chance handler implementation. This packet definition exists to ensure that the packet gets reflected in the documentation and ground system definitions.
- **Event_Packetizer_Instance.Events_Packet** - This packet contains events as subpackets.

2.10 Faults

The table below shows the faults for the Pico Example assembly.

Table 9: Pico Example Faults

Fault ID	Fault Name	Parameter Type
0x0001 (1)	Task_Watchdog_Instance.Slow_Rate_Group_Fault	Packed_Connector_Index.T
0x0002 (2)	Task_Watchdog_Instance.Fast_Rate_Group_Fault	Packed_Connector_Index.T
0x0003 (3)	Fault_Producer_Instance.Fault_1	-
0x0004 (4)	Fault_Producer_Instance.Fault_2	Packed_Natural.T

Fault Descriptions:

- **Task_Watchdog_Instance.Slow_Rate_Group_Fault** - Slow rate group monitoring.
- **Task_Watchdog_Instance.Fast_Rate_Group_Fault** - Fast rate group monitoring.
- **Fault_Producer_Instance.Fault_1** - First fault that the component can send.
- **Fault_Producer_Instance.Fault_2** - Second fault that the component can send.

3 Appendix

3.1 Connections

Table 10: Pico Example Connections

Number	From	To	Kind
1	Ticker_Instance.Sys_Time_T_Get	System_Time_Instance.Sys_Time_T_Return	get-return
2	Tick_Divider_Instance.Sys_Time_T_Get	System_Time_Instance.Sys_Time_T_Return	get-return
3	Slow_Rate_Group.Sys_Time_T_Get	System_Time_Instance.Sys_Time_T_Return	get-return
4	Fast_Rate_Group.Sys_Time_T_Get	System_Time_Instance.Sys_Time_T_Return	get-return

5	Watchdog_Rate_Group. Sys_Time_T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
6	Command_Router_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
7	Counter_Instance.Sys_ Time_T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
8	Oscillator_A.Sys_ Time_T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
9	Oscillator_B.Sys_ Time_T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
10	Event_Packetizer_ Instance.Sys_Time_T_ Get	System_Time_Instance. Sys_Time_T_Return	get-return
11	Product_Database_ Instance.Sys_Time_T_ Get	System_Time_Instance. Sys_Time_T_Return	get-return
12	Product_Packetizer_ Instance.Sys_Time_T_ Get	System_Time_Instance. Sys_Time_T_Return	get-return
13	Ccsds_Command_ Depacketizer_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
14	Cpu_Monitor_Instance. Sys_Time_T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
15	Stack_Monitor_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
16	Queue_Monitor_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
17	Event_Filter_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
18	Event_Limiter_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return

19	Zero_Divider_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
20	Task_Watchdog_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
21	Fault_Correction_ Instance.Sys_Time_T_ Get	System_Time_Instance. Sys_Time_T_Return	get-return
22	Fault_Producer_ Instance.Sys_Time_ T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
23	Ccsds_Serial_ Interface_Instance. Sys_Time_T_Get	System_Time_Instance. Sys_Time_T_Return	get-return
24	Adc_Data_Collector_ Instance.Sys_Time_T_ Get	System_Time_Instance. Sys_Time_T_Return	get-return
25	Ticker_Instance.Tick_ T_Send	Tick_Divider_ Instance.Tick_T_Recv_ Sync	send-recv_sync
26	Tick_Divider_ Instance.Tick_T_Send [1]	Watchdog_Rate_Group. Tick_T_Recv_Async	send-recv_async
27	Tick_Divider_ Instance.Tick_T_Send [2]	Slow_Rate_Group.Tick_ T_Recv_Async	send-recv_async
28	Tick_Divider_ Instance.Tick_T_Send [3]	Fast_Rate_Group.Tick_ T_Recv_Async	send-recv_async
29	Slow_Rate_Group.Tick_ T_Send [1]	Counter_Instance. Tick_T_Recv_Sync	send-recv_sync
30	Slow_Rate_Group.Tick_ T_Send [2]	Event_Packetizer_ Instance.Tick_T_Recv_ Sync	send-recv_sync
31	Slow_Rate_Group.Tick_ T_Send [3]	Cpu_Monitor_Instance. Tick_T_Recv_Sync	send-recv_sync

32	Slow_Rate_Group.Tick_T_Send [4]	Queue_Monitor_Instance.Tick_T_Recv_Sync	send-recv_sync
33	Slow_Rate_Group.Tick_T_Send [5]	Stack_Monitor_Instance.Tick_T_Recv_Sync	send-recv_sync
34	Slow_Rate_Group.Tick_T_Send [6]	Event_Filter_Instance.Tick_T_Recv_Sync	send-recv_sync
35	Slow_Rate_Group.Tick_T_Send [7]	Event_Limiter_Instance.Tick_T_Recv_Sync	send-recv_sync
36	Slow_Rate_Group.Tick_T_Send [8]	Adc_Data_Collector_Instance.Tick_T_Recv_Sync	send-recv_sync
37	Fast_Rate_Group.Tick_T_Send [1]	Oscillator_A.Tick_T_Recv_Sync	send-recv_sync
38	Fast_Rate_Group.Tick_T_Send [2]	Oscillator_B.Tick_T_Recv_Sync	send-recv_sync
39	Fast_Rate_Group.Tick_T_Send [3]	Product_Packetizer_Instance.Tick_T_Recv_Sync	send-recv_sync
40	Watchdog_Rate_Group.Tick_T_Send	Task_Watchdog_Instance.Tick_T_Recv_Sync	send-recv_sync
41	Tick_Divider_Instance.Event_T_Send	Event_Filter_Instance.Event_T_Recv_Sync	send-recv_sync
42	Slow_Rate_Group.Event_T_Send	Event_Filter_Instance.Event_T_Recv_Sync	send-recv_sync
43	Fast_Rate_Group.Event_T_Send	Event_Filter_Instance.Event_T_Recv_Sync	send-recv_sync
44	Watchdog_Rate_Group.Event_T_Send	Event_Filter_Instance.Event_T_Recv_Sync	send-recv_sync

45	Command_Router_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
46	Counter_Instance. Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
47	Oscillator_A.Event_T_ Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
48	Oscillator_B.Event_T_ Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
49	Product_Packetizer_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
50	Stack_Monitor_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
51	Queue_Monitor_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
52	Cpu_Monitor_Instance. Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
53	Event_Filter_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
54	Event_Limiter_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
55	Zero_Divider_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
56	Task_Watchdog_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
57	Fault_Correction_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync

58	Ccsds_Serial_ Interface_Instance. Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
59	Event_Filter_ Instance.Event_ Forward_T_Send	Event_Limiter_ Instance.Event_T_ Recv_Sync	send-recv_sync
60	Event_Limiter_ Instance.Event_ Forward_T_Send	Event_Packetizer_ Instance.Event_T_ Recv_Sync	send-recv_sync
61	Product_Database_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
62	Ccsds_Command_ Depacketizer_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
63	Fault_Producer_ Instance.Event_T_Send	Event_Filter_ Instance.Event_T_ Recv_Sync	send-recv_sync
64	Command_Router_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
65	Counter_Instance. Command_Response_T_ Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
66	Oscillator_A.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
67	Oscillator_B.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
68	Product_Packetizer_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
69	Event_Packetizer_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
70	Ccsds_Command_ Depacketizer_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async

71	Product_Database_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
72	Stack_Monitor_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
73	Queue_Monitor_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
74	Cpu_Monitor_Instance. Command_Response_T_ Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
75	Event_Filter_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
76	Event_Limiter_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
77	Task_Watchdog_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
78	Fault_Correction_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
79	Zero_Divider_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
80	Fault_Producer_ Instance.Command_ Response_T_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
81	Ccsds_Command_ Depacketizer_ Instance.Command_ T_Send	Command_Router_ Instance.Command_ T_To_Route_Recv_Async	send-recv_async
82	Command_Router_ Instance.Command_ T_Send [1]	Command_Router_ Instance.Command_ T_Recv_Async	send-recv_async
83	Command_Router_ Instance.Command_ T_Send [2]	Counter_Instance. Command_T_Recv_Async	send-recv_async

84	Command_Router_ Instance.Command_ T_Send [3]	Oscillator_A.Command_ T_Recv_Async	send-recv_async
85	Command_Router_ Instance.Command_ T_Send [4]	Oscillator_B.Command_ T_Recv_Async	send-recv_async
86	Command_Router_ Instance.Command_ T_Send [5]	Product_Packetizer_ Instance.Command_T_ Recv_Async	send-recv_async
87	Command_Router_ Instance.Command_ T_Send [6]	Event_Packetizer_ Instance.Command_T_ Recv_Sync	send-recv_sync
88	Command_Router_ Instance.Command_ T_Send [7]	Product_Database_ Instance.Command_T_ Recv_Sync	send-recv_sync
89	Command_Router_ Instance.Command_ T_Send [8]	Ccsds_Command_ Depacketizer_ Instance.Command_ T_Recv_Sync	send-recv_sync
90	Command_Router_ Instance.Command_ T_Send [9]	Stack_Monitor_ Instance.Command_ T_Recv_Sync	send-recv_sync
91	Command_Router_ Instance.Command_ T_Send [10]	Queue_Monitor_ Instance.Command_ T_Recv_Sync	send-recv_sync
92	Command_Router_ Instance.Command_ T_Send [11]	Cpu_Monitor_Instance. Command_T_Recv_Sync	send-recv_sync
93	Command_Router_ Instance.Command_ T_Send [12]	Event_Filter_ Instance.Command_ T_Recv_Sync	send-recv_sync
94	Command_Router_ Instance.Command_ T_Send [13]	Event_Limiter_ Instance.Command_ T_Recv_Sync	send-recv_sync
95	Command_Router_ Instance.Command_ T_Send [14]	Zero_Divider_ Instance.Command_ T_Recv_Sync	send-recv_sync
96	Command_Router_ Instance.Command_ T_Send [15]	Fault_Correction_ Instance.Command_T_ Recv_Async	send-recv_async

97	Command_Router_ Instance.Command_ T_Send [16]	Task_Watchdog_ Instance.Command_ T_Recv_Sync	send-recv_sync
98	Command_Router_ Instance.Command_ T_Send [17]	Fault_Producer_ Instance.Command_ T_Recv_Sync	send-recv_sync
99	Command_Router_ Instance.Command_ Response_T_To_ Forward_Send	Command_Router_ Instance.Command_ Response_T_Recv_Async	send-recv_async
100	Oscillator_A.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
101	Oscillator_B.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
102	Event_Packetizer_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
103	Ccsds_Command_ Depacketizer_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
104	Slow_Rate_Group.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
105	Fast_Rate_Group.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
106	Watchdog_Rate_Group. Data_Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
107	Command_Router_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
108	Product_Database_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync

109	Stack_Monitor_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
110	Queue_Monitor_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
111	Cpu_Monitor_Instance. Data_Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
112	Event_Filter_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
113	Event_Limiter_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
114	Task_Watchdog_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
115	Fault_Correction_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
116	Adc_Data_Collector_ Instance.Data_ Product_T_Send	Product_Database_ Instance.Data_ Product_T_Recv_Sync	send-recv_sync
117	Product_Packetizer_ Instance.Data_ Product_Fetch_T_ Request	Product_Database_ Instance.Data_ Product_Fetch_T_ Service	request-service
118	Event_Packetizer_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
119	Product_Packetizer_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
120	Counter_Instance. Packet_T_Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync

121	Ccsds_Command_ Depacketizer_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
122	Product_Database_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
123	Cpu_Monitor_Instance. Packet_T_Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
124	Queue_Monitor_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
125	Stack_Monitor_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
126	Event_Filter_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
127	Event_Limiter_ Instance.Packet_T_ Send	Ccsds_Packetizer_ Instance.Packet_T_ Recv_Sync	send-recv_sync
128	Slow_Rate_Group.Pet_ T_Send	Task_Watchdog_ Instance.Pet_T_Recv_ Sync	send-recv_sync
129	Fast_Rate_Group.Pet_ T_Send	Task_Watchdog_ Instance.Pet_T_Recv_ Sync	send-recv_sync
130	Task_Watchdog_ Instance.Fault_T_Send	Fault_Correction_ Instance.Fault_T_ Recv_Async	send-recv_async
131	Fault_Producer_ Instance.Fault_T_Send	Fault_Correction_ Instance.Fault_T_ Recv_Async	send-recv_async
132	Fault_Correction_ Instance.Command_T_ Send	Command_Router_ Instance.Command_ T_To_Route_Recv_Sync	send-recv_sync

133	Ccsds_Packetizer_Instance.Ccsds_Space_Packet_T_Send	Ccsds_Serial_Interface_Instance.Ccsds_Space_Packet_T_Recv_Async	send-recv_async
134	Ccsds_Serial_Interface_Instance.Ccsds_Space_Packet_T_Send	Ccsds_Command_Depacketizer_Instance.Ccsds_Space_Packet_T_Recv_Sync	send-recv_sync

Connection Descriptions:

- **Ticker_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Tick_Divider_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Slow_Rate_Group.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Fast_Rate_Group.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Watchdog_Rate_Group.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Command_Router_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Counter_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Oscillator_A.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Oscillator_B.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Event_Packetizer_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Product_Database_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Product_Packetizer_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Ccsds_Command_Depacketizer_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Cpu_Monitor_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Stack_Monitor_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Queue_Monitor_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Event_Filter_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Event_Limiter_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*

- **Zero_Divider_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Task_Watchdog_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Fault_Correction_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Fault_Producer_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Ccsds_Serial_Interface_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Adc_Data_Collector_Instance.Sys_Time_T_Get-System_Time_Instance.Sys_Time_T_Return** - *No description provided.*
- **Ticker_Instance.Tick_T_Send-Tick_Divider_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Tick_Divider_Instance.Tick_T_Send[1]-Watchdog_Rate_Group.Tick_T_Recv_Async** - *No description provided.*
- **Tick_Divider_Instance.Tick_T_Send[2]-Slow_Rate_Group.Tick_T_Recv_Async** - *No description provided.*
- **Tick_Divider_Instance.Tick_T_Send[3]-Fast_Rate_Group.Tick_T_Recv_Async** - *No description provided.*
- **Slow_Rate_Group.Tick_T_Send[1]-Counter_Instance.Tick_T_Recv_Sync** - Schedule connection from the rate group to the counter
- **Slow_Rate_Group.Tick_T_Send[2]-Event_Packetizer_Instance.Tick_T_Recv_Sync** - Schedule connection from the rate group to the packetizer
- **Slow_Rate_Group.Tick_T_Send[3]-Cpu_Monitor_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Tick_T_Send[4]-Queue_Monitor_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Tick_T_Send[5]-Stack_Monitor_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Tick_T_Send[6]-Event_Filter_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Tick_T_Send[7]-Event_Limiter_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Tick_T_Send[8]-Adc_Data_Collector_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Fast_Rate_Group.Tick_T_Send[1]-Oscillator_A.Tick_T_Recv_Sync** - Schedule connection from the rate group to the oscillator
- **Fast_Rate_Group.Tick_T_Send[2]-Oscillator_B.Tick_T_Recv_Sync** - Schedule connection from the rate group to the oscillator
- **Fast_Rate_Group.Tick_T_Send[3]-Product_Packetizer_Instance.Tick_T_Recv_Sync** - Schedule connection from the rate group to the packetizer
- **Watchdog_Rate_Group.Tick_T_Send-Task_Watchdog_Instance.Tick_T_Recv_Sync** - *No description provided.*
- **Tick_Divider_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync**

- *No description provided.*

- **Fast_Rate_Group.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Watchdog_Rate_Group.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Counter_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Oscillator_A.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Oscillator_B.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Product_Packetizer_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Stack_Monitor_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Queue_Monitor_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Cpu_Monitor_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Event_Filter_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Event_Limiter_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Zero_Divider_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Task_Watchdog_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Fault_Correction_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Ccsds_Serial_Interface_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Event_Filter_Instance.Event_Forward_T_Send-Event_Limiter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Event_Limiter_Instance.Event_Forward_T_Send-Event_Packetizer_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Product_Database_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Ccsds_Command_Depacketizer_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Fault_Producer_Instance.Event_T_Send-Event_Filter_Instance.Event_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Counter_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*

- **Oscillator_A.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Oscillator_B.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Product_Packetizer_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Event_Packetizer_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Ccsds_Command_Depacketizer_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Product_Database_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Stack_Monitor_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Queue_Monitor_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Cpu_Monitor_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Event_Filter_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Event_Limiter_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Task_Watchdog_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Fault_Correction_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Zero_Divider_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Fault_Producer_Instance.Command_Response_T_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Ccsds_Command_Depacketizer_Instance.Command_T_Send-Command_Router_Instance.Command_T_To_Route_Recv_Async** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[1]-Command_Router_Instance.Command_T_Recv_Async** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[2]-Counter_Instance.Command_T_Recv_Async** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[3]-Oscillator_A.Command_T_Recv_Async** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[4]-Oscillator_B.Command_T_Recv_Async** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[5]-Product_Packetizer_Instance.Command_T_Recv_Async** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[6]-Event_Packetizer_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[7]-Product_Database_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[8]-Ccsds_Command_**

Depacketizer_Instance.Command_T_Recv_Sync - *No description provided.*

- **Command_Router_Instance.Command_T_Send[9]-Stack_Monitor_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[10]-Queue_Monitor_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[11]-Cpu_Monitor_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[12]-Event_Filter_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[13]-Event_Limiter_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[14]-Zero_Divider_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[15]-Fault_Correction_Instance.Command_T_Recv_Async** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[16]-Task_Watchdog_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_T_Send[17]-Fault_Producer_Instance.Command_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Command_Response_T_To_Forward_Send-Command_Router_Instance.Command_Response_T_Recv_Async** - *No description provided.*
- **Oscillator_A.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - Data product connection between the oscillator and packetizer
- **Oscillator_B.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - Data product connection between the oscillator and packetizer
- **Event_Packetizer_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Ccsds_Command_Depacketizer_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Fast_Rate_Group.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Watchdog_Rate_Group.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Command_Router_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Product_Database_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Stack_Monitor_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Queue_Monitor_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Cpu_Monitor_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Event_Filter_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*

- **Event_Limiter_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Task_Watchdog_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Fault_Correction_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Adc_Data_Collector_Instance.Data_Product_T_Send-Product_Database_Instance.Data_Product_T_Recv_Sync** - *No description provided.*
- **Product_Packetizer_Instance.Data_Product_Fetch_T_Request-Product_Database_Instance.Data_Product_Fetch_T_Service** - *No description provided.*
- **Event_Packetizer_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Product_Packetizer_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Counter_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Ccsds_Command_Depacketizer_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Product_Database_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Cpu_Monitor_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Queue_Monitor_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Stack_Monitor_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Event_Filter_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Event_Limiter_Instance.Packet_T_Send-Ccsds_Packetizer_Instance.Packet_T_Recv_Sync** - *No description provided.*
- **Slow_Rate_Group.Pet_T_Send-Task_Watchdog_Instance.Pet_T_Recv_Sync[1]** - *No description provided.*
- **Fast_Rate_Group.Pet_T_Send-Task_Watchdog_Instance.Pet_T_Recv_Sync[2]** - *No description provided.*
- **Task_Watchdog_Instance.Fault_T_Send-Fault_Correction_Instance.Fault_T_Recv_Async** - *No description provided.*
- **Fault_Producer_Instance.Fault_T_Send-Fault_Correction_Instance.Fault_T_Recv_Async** - *No description provided.*
- **Fault_Correction_Instance.Command_T_Send-Command_Router_Instance.Command_T_To_Route_Recv_Sync** - We connect the fault correction command response to the command router synchronous connector for the fastest, most reliable execution. This bypasses the command router's queue.
- **Ccsds_Packetizer_Instance.Ccsds_Space_Packet_T_Send-Ccsds_Serial_Interface_Instance.Ccsds_Space_Packet_T_Recv_Async** - *No description provided.*
- **Ccsds_Serial_Interface_Instance.Ccsds_Space_Packet_T_Send-Ccsds_Command_Depacketizer_Instance.Ccsds_Space_Packet_T_Recv_Sync** - *No description provided.*

3.2 Packed Types

The following section outlines any complex data types used in the assembly in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, or event parameters.

Ccsds_Command_Header.T:

Record for a LASP-specific CCSDS command header.

Table 11: Ccsds_Command_Header Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Primary_Header	Ccsds_Primary_Header.T	-	48	0	47
Secondary_Header	Ccsds_Command_Secondary_Header.T	-	16	48	63

Field Descriptions:

- **Primary_Header** - The CCSDS primary header
- **Secondary_Header** - The command secondary header

Ccsds_Command_Secondary_Header.T:

Record for the LASP-specific command secondary header. *Preamble (inline Ada definitions):*

```
type Function_Code_Type is mod 2**7;
type One_Bit_Pad_Type is mod 2**1;
```

Table 12: Ccsds_Command_Secondary_Header Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Reserved	One_Bit_Pad_Type	0 to 1	1	0	0
Function_Code	Function_Code_Type	0 to 127	7	1	7
Checksum	Interfaces.Unsigned_8	0 to 255	8	8	15

Field Descriptions:

- **Reserved** - Reserve bit.
- **Function_Code** - The command function code.
- **Checksum** - An 8 bit checksum over the entire command packet

Ccsds_Primary_Header.T:

Record for the CCSDS Packet Primary Header *Preamble (inline Ada definitions):*

```
subtype Three_Bit_Version_Type is Interfaces.Unsigned_8 range 0 .. 7;
type Ccsds_Apid_Type is mod 2**11;
type Ccsds_Sequence_Count_Type is mod 2**14;
```

Table 13: Ccsds_Primary_Header Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Version	Three_Bit_Version_Type	0 to 7	3	0	2
Packet_Type	Ccsds_Enums.Ccsds_Packet_Type.E	0 => Telemetry 1 => Telecommand	1	3	3
Secondary_Header	Ccsds_Enums.Ccsds_Secondary_Header_Indicator.E	0 => Secondary_Header_Not_Present 1 => Secondary_Header_Present	1	4	4
Apid	Ccsds_Apid_Type	0 to 2047	11	5	15
Sequence_Flag	Ccsds_Enums.Ccsds_Sequence_Flag.E	0 => Continuationsegment 1 => Firstsegment 2 => Lastsegment 3 => Unsegmented	2	16	17
Sequence_Count	Ccsds_Sequence_Count_Type	0 to 16383	14	18	31
Packet_Length	Interfaces.Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Version** - Packet Version Number
- **Packet_Type** - Packet Type
- **Secondary_Header** - Does packet have CCSDS secondary header
- **Apid** - Application process identifier
- **Sequence_Flag** - Sequence Flag
- **Sequence_Count** - Packet Sequence Count
- **Packet_Length** - This is the packet data length. One added to this number corresponds to the number of bytes included in the data section of the CCSDS Space Packet.

Ccsds_Space_Packet.T:

Record for the CCSDS Space Packet *Preamble (inline Ada definitions):*

```
use Basic_Types;
subtype Ccsds_Data_Type is Byte_Array (0 ..
↪  Configuration.Ccsds_Packet_Buffer_Size - 1);
```

Table 14: Ccsds_Space_Packet Packed Record : 10240 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Ccsds_ Primary_ Header.T	-	48	0	47	–
Data	Ccsds_Data_ Type	-	10192	48	10239	Header. Packet_Length

Field Descriptions:

- **Header** - The CCSDS Primary Header
- **Data** - User Data Field

Command.T:

Generic command packet for holding arbitrary commands

Table 15: Command Packed Record : 2080 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Command_ Header.T	-	40	0	39	–
Arg_Buffer	Command_ Types. Command_Arg_ Buffer_Type	-	2040	40	2079	Header.Arg_ Buffer_Length

Field Descriptions:

- **Header** - The command header
- **Arg_Buffer** - A buffer that contains the command arguments

Command_Header.T:

Generic command header for holding arbitrary commands

Table 16: Command_Header Packed Record : 40 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Source_Id	Command_Types. Command_Source_Id	0 to 65535	16	0	15
Id	Command_Types. Command_Id	0 to 65535	16	16	31
Arg_Buffer_Length	Command_Types. Command_Arg_Buffer_ Length_Type	0 to 255	8	32	39

Field Descriptions:

- **`Source_Id`** - The source ID. An ID assigned to a command sending component.
- **`Id`** - The command identifier
- **`Arg_Buffer_Length`** - The number of bytes used in the command argument buffer

Command_Id.T:

A packed record which holds a command identifier.

Table 17: Command_Id Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Command_Types. Command_Id	0 to 65535	16	0	15

Field Descriptions:
- **`Id`** - The command identifier

Command_Id_Status.T:

Record for holding a command identifier and command response status.

Table 18: Command_Id_Status Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Command_Types. Command_Id	0 to 65535	16	0	15
Status	Command_Enums. Command_ Response_ Status.E	0 => Success 1 => Failure 2 => Id_Error 3 => Validation_Error 4 => Length_Error 5 => Dropped 6 => Register 7 => Register_Source	8	16	23

Field Descriptions:
- **`Id`** - The command ID for the command response.
- **`Status`** - The command execution status.

Command_Response.T:

Record for holding command response data.

Table 19: Command_Response Packed Record : 56 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Source_Id	Command_ Types.Command_ Source_Id	0 to 65535	16	0	15

Registration_ Id	Command_ Types.Command_ Registration_ Id	0 to 65535	16	16	31
Command_Id	Command_Types. Command_Id	0 to 65535	16	32	47
Status	Command_Enums. Command_ Response_ Status.E	0 => Success 1 => Failure 2 => Id_Error 3 => Validation_Error 4 => Length_Error 5 => Dropped 6 => Register 7 => Register_Source	8	48	55

Field Descriptions:

- **Source_Id** - The source ID. An ID assigned to a command sending component.
- **Registration_Id** - The registration ID. An ID assigned to each registered component at initialization.
- **Command_Id** - The command ID for the command response.
- **Status** - The command execution status.

Command_Router_Arg.T:

A 32-bit unsigned integer with range 0 to 999. *Preamble (inline Ada definitions):*

```
subtype Value_Type is Natural range 0 .. 999;
```

Table 20: Command_Router_Arg Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Value_Type	0 to 999	32	0	31

Field Descriptions:

- **Value** - The 32-bit unsigned integer with range 0 to 999.

Cycle_Slip_Param.T:

This is a type that contains useful information about a cycle slip.

Table 21: Cycle_Slip_Param Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Slipped_Tick	Tick.T	-	96	0	95
Num_Slips	Interfaces. Unsigned_16	0 to 65535	16	96	111

Field Descriptions:

- **Slipped_Tick** - The tick during which the cycle slip occurred.
- **Num_Slips** - The number of cycle slips that have occurred.

Data_Product.T:

Generic data product packet for holding arbitrary data types

Table 22: Data_Product Packed Record : 344 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Data_Product_Header.T	-	88	0	87	–
Buffer	Data_Product_Types.Data_Product_Buffer_Type	-	256	88	343	Header.Buffer_Length

Field Descriptions:
- **Header** - The data product header
- **Buffer** - A buffer that contains the data product type

Data_Product_Fetch.T:

A packed record which holds information for a data product request.

Table 23: Data_Product_Fetch Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types.Data_Product_Id	0 to 65535	16	0	15

Field Descriptions:
- **Id** - The data product identifier

Data_Product_Header.T:

Generic data_product packet for holding arbitrary data_product types

Table 24: Data_Product_Header Packed Record : 88 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	64	0	63
Id	Data_Product_Types.Data_Product_Id	0 to 65535	16	64	79
Buffer_Length	Data_Product_Types.Data_Product_Buffer_Length_Type	0 to 32	8	80	87

Field Descriptions:
- **Time** - The timestamp for the data product item.
- **Id** - The data product identifier

- **Buffer_Length** - The number of bytes used in the data product buffer

Data_Product_Id.T:

A packed record which holds a data product identifier.

Table 25: Data_Product_Id Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types.Data_Product_Id	0 to 65535	16	0	15

Field Descriptions:
- **Id** - The data product identifier

Data_Product_Poly_Event.T:

Data product with 4 byte data buffer.

Table 26: Data_Product_Poly_Event Packed Record : 120 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Header	Data_Product_Header.T	-	88	0	87
Data	Basic_Types.Poly_32_Type	-	32	88	119

Field Descriptions:
- **Header** - The data product header
- **Data** - The polymorphic type.

Data_Product_Poly_Extract.T:

Contains information to extract a poly type from a data product. *Preamble (inline Ada definitions):*

```
subtype Data_Product_Bit_Offset_Type is Natural range Natural'First ..
↪  Data_Product_Types.Data_Product_Buffer_Type'Length *
↪  Basic_Types.Byte'Object_Size; subtype Poly_Type_Size_Type is Positive range
↪  Positive'First .. 32;
```

Table 27: Data_Product_Poly_Extract Packed Record : 40 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types.Data_Product_Id	0 to 65535	16	0	15
Offset	Data_Product_Bit_Offset_Type	0 to 256	16	16	31
Size	Poly_Type_Size_Type	1 to 32	8	32	39

Field Descriptions:

- **Id** - ID of the data product.
- **Offset** - Offset of the data product item (in bits).
- **Size** - Size of the data product item (in bits).

Data_Product_Poly_Type.T:

Data product poly type, for dumping arbitrary data products.

Table 28: Data_Product_Poly_Type Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	64	0	63
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	64	79
Data	Basic_Types.Poly_ 32_Type	-	32	80	111

Field Descriptions:
- **Time** - The timestamp for the data product item.
- **Id** - The data product identifier
- **Data** - The polymorphic type.

Data_Product_Return.T:

This record holds data returned from a data product fetch request.

Table 29: Data_Product_Return Packed Record : 352 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
The_ Status	Data_ Product_ Enums. Fetch_ Status.E	0 => Success 1 => Not_Available 2 => Id_Out_Of_Range	8	0	7	–
The_Data_ Product	Data_ Product.T	-	344	8	351	–

Field Descriptions:
- **The_Status** - A status relating whether or not the data product fetch was successful or not.
- **The_Data_Product** - The data product item returned.

Delta_Time.T:

A record which holds a time difference using GPS format including seconds and subseconds.

Table 30: Delta_Time Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit

Seconds	Interfaces. Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces. Unsigned_32	0 to 4294967295	32	32	63

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of 1/(2^32) sub-seconds.

Event.T:

Generic event packet for holding arbitrary events

Table 31: Event Packed Record : 344 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Event_Header.T	-	88	0	87	–
Param_Buffer	Event_Types. Parameter_ Buffer_Type	-	256	88	343	Header.Param_ Buffer_Length

Field Descriptions:

- **Header** - The event header
- **Param_Buffer** - A buffer that contains the event parameters

Event_Component_State_Type.T:

This record is for the data product that indicates if the event filter component is enabled for filtering or disabled all together.

Table 32: Event_Component_State_Type Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Component_Filter_ State	Event_Filter_ Entry_Enums. Global_Filter_ State.E	0 => Disabled 1 => Enabled	8	0	7

Field Descriptions:

- **Component_Filter_State** - Flag to indicate if the component is enabled or disabled at a overriding level from the individual event states

Event_Enable_State_Type.T:

This record contains the definition for a two event ID type for ranges in the event limiter commands as well as an issue packet type for issuing packets

Table 33: Event_Enable_State_Type Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Component_Enable_State	Two_Counter_Entry_Enums.Event_State_Type.E	0 => Disabled 1 => Enabled	8	0	7

Field Descriptions:

- **Component_Enable_State** - Flag to indicate if the component is enabled or disabled at a overriding level

Event_Filter_Id_Range.T:

This record contains the definition for a two event ID type for ranges in the event filter commands as well as an issue packet type for issuing packets

Table 34: Event_Filter_Id_Range Packed Record : 40 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Start_Event_Id	Event_Id.T	-	16	0	15
Stop_Event_Id	Event_Id.T	-	16	16	31
Issue_State_Packet	Event_Filter_Enums.Issue_Packet_Type.E	0 => No_Issue 1 => Issue	8	32	39

Field Descriptions:

- **Start_Event_Id** - The starting event ID to begin the range
- **Stop_Event_Id** - The last event ID to end the range
- **Issue_State_Packet** - Flag to indicate if we dump the states after the change is complete

Event_Filter_Single_Event_Cmd_Type.T:

This record contains the definition for a single event ID in the event filter commands as well as an issue packet type for issuing packets

Table 35: Event_Filter_Single_Event_Cmd_Type Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Event_To_Update	Event_Id.T	-	16	0	15
Issue_State_Packet	Event_Filter_Enums.Issue_Packet_Type.E	0 => No_Issue 1 => Issue	8	16	23

Field Descriptions:

- **Event_To_Update** - The event ID to update
- **Issue_State_Packet** - Flag to indicate if we dump the states after the change is complete

Event_Header.T:

Generic event packet for holding arbitrary events

Table 36: Event_Header Packed Record : 88 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	64	0	63
Id	Event_Types.Event_Id	0 to 65535	16	64	79
Param_Buffer_Length	Event_Types.Parameter_Buffer_Length_Type	0 to 32	8	80	87

Field Descriptions:

- **Time** - The timestamp for the event.
- **Id** - The event identifier
- **Param_Buffer_Length** - The number of bytes used in the param buffer

Event_Id.T:

A packed record which holds an event identifier.

Table 37: Event_Id Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Event_Types.Event_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The event identifier

Event_Id_Array.T:

Packed array of events so that it can be used with the event for which ids were limited

Table 38: Event_Id_Array Packed Array : 160 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Event_Types.Event_Id	0 to 65535	16	10	160

Event_Limiter_Id_Range.T:

This record contains the definition for a two event ID type for ranges in the event limiter commands as well as an issue packet type for issuing packets

Table 39: Event_Limiter_Id_Range Packed Record : 40 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Start_Event_Id	Event_Id.T	-	16	0	15
Stop_Event_Id	Event_Id.T	-	16	16	31

Issue_State_Packet	Event_Limiter_ Enums.Issue_ Packet_Type.E	0 => No_Issue 1 => Issue	8	32	39

Field Descriptions:

- **Start_Event_Id** - The starting event ID to begin the range
- **Stop_Event_Id** - The last event ID to end the range
- **Issue_State_Packet** - Flag to indicate if we dump the states after the change is complete

Event_Limiter_Num_Events_Type.T:

This record contains the definition for the format of the event for this component that contains up to 10 event IDs of those that were limited.

Table 40: Event_Limiter_Num_Events_Type Packed Record : 184 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Num_Events_Limited	Interfaces. Unsigned_16	0 to 65535	16	0	15
Num_Event_Ids	Interfaces. Unsigned_8	0 to 255	8	16	23
Event_Id_Limited_ Array	Event_Id_Array.T	-	160	24	183

Field Descriptions:

- **Num_Events_Limited** - The number of events limited since last issuing the event.
- **Num_Event_Ids** - The number of event IDs contained in the event that indicate that event was limited.
- **Event_Id_Limited_Array** - The Array that list the event IDs that were limited since the last event message.

Event_Limiter_Persistence_Type.T:

This record contains the definition for a packed persistence type

Table 41: Event_Limiter_Persistence_Type Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Persistence	Two_Counter_Entry. Persistence_Type	1 to 7	8	0	7

Field Descriptions:

- **Persistence** - The persistence that is used when limiting events.

Event_Single_State_Cmd_Type.T:

This record contains the definition for a two event ID type for ranges in the event limiter commands as well as an issue packet type for issuing packets

Table 42: Event_Single_State_Cmd_Type Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Event_To_Update	Event_Id.T	-	16	0	15
Issue_State_Packet	Event_Limiter_ Enums.Issue_ Packet_Type.E	0 => No_Issue 1 => Issue	8	16	23

Field Descriptions:

- **Event_To_Update** - The starting event ID to begin the range
- **Issue_State_Packet** - Flag to indicate if we dump the states after the change is complete

Fault.T:

Generic fault packet for holding arbitrary faults.

Table 43: Fault Packed Record : 152 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Fault_Header.T	-	88	0	87	–
Param_Buffer	Fault_Types. Parameter_ Buffer_Type	-	64	88	151	Header.Param_ Buffer_Length

Field Descriptions:

- **Header** - The fault header
- **Param_Buffer** - A buffer that contains the fault parameters

Fault_Header.T:

Generic fault header.

Table 44: Fault_Header Packed Record : 88 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	64	0	63
Id	Fault_Types.Fault_ Id	0 to 65535	16	64	79
Param_Buffer_Length	Fault_Types. Parameter_Buffer_ Length_Type	0 to 8	8	80	87

Field Descriptions:

- **Time** - The timestamp for the fault.
- **Id** - The fault identifier
- **Param_Buffer_Length** - The number of bytes used in the param buffer

Fault_Static.T:

Generic fault packet for holding arbitrary faults. This is the same as the Fault.T type, except that it is not variable sized, it is always maximum sized. This can be useful for sending events with faults in them.

Table 45: Fault_Static Packed Record : 152 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Header	Fault_Header.T	-	88	0	87
Param_Buffer	Fault_Types.Parameter_ Buffer_Type	-	64	88	151

Field Descriptions:

- **Header** - The fault header
- **Param_Buffer** - A buffer that contains the fault parameters

Full_Queue_Param.T:

This is a type that contains useful information about a component full queue.

Table 46: Full_Queue_Param Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Dropped_Tick	Tick.T	-	96	0	95
Index	Connector_Types. Connector_Index_ Type	1 to 65535	16	96	111

Field Descriptions:

- **Dropped_Tick** - The tick during which the component's queue was found to be full.
- **Index** - The rate group index number of the component that had a full queue.

Invalid_Command_Info.T:

Record for holding information about an invalid command

Table 47: Invalid_Command_Info Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Command_Types. Command_Id	0 to 65535	16	0	15
Errant_Field_ Number	Interfaces. Unsigned_32	0 to 4294967295	32	16	47
Errant_Field	Basic_Types.Poly_ Type	-	64	48	111

Field Descriptions:

- **Id** - The command Id received.
- **Errant_Field_Number** - The field that was invalid. 1 is the first field, 0 means unknown field, 2**32 means that the length field of the command was invalid.
- **Errant_Field** - A polymorphic type containing the bad field data, or length when Errant_Field_Number is 2**32.

Invalid_Data_Product_Length.T:

A packed record which holds data related to an invalid data product length.

Table 48: Invalid_Data_Product_Length Packed Record : 96 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Header	Data_Product_Header.T	-	88	0	87
Expected_Length	Data_Product_Types. Data_Product_Buffer_ Length_Type	0 to 32	8	88	95

Field Descriptions:

- **Header** - The packet identifier
- **Expected_Length** - The packet length bound that the length failed to meet.

Invalid_Packet_Id.T:

A packed record which holds a packet identifier and data product identifier

Table 49: Invalid_Packet_Id Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Packet_Id	Packet_Types. Packet_Id	0 to 65535	16	0	15
Command_Id	Command_Types. Command_Id	0 to 65535	16	16	31

Field Descriptions:

- **Packet_Id** - The packet identifier
- **Command_Id** - The command id

Invalid_Packet_Length.T:

A packed record which holds data related to an invalid command packet length.

Table 50: Invalid_Packet_Length Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Ccsds_Header	Ccsds_ Primary_ Header.T	-	48	0	47
Length	Integer	-2147483648 to 2147483647	32	48	79
Length_Bound	Integer	-2147483648 to 2147483647	32	80	111

Field Descriptions:

- **Ccsds_Header** - The packet identifier
- **Length** - The packet length
- **Length_Bound** - The packet length bound that the length failed to meet.

Invalid_Packet_Xor8_Info.T:

A packed record which holds data related to an invalid checksummed CCSDS command packet.

Table 51: Invalid_Packet_Xor8_Info Packed Record : 80 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Ccsds_Header	Ccsds_Command_ Header.T	-	64	0	63
Computed_Checksum	Xor_8.Xor_8_Type	0 to 255	8	64	71
Expected_Checksum	Xor_8.Xor_8_Type	0 to 255	8	72	79

Field Descriptions:

- **Ccsds_Header** - The CCSDS command header.
- **Computed_Checksum** - The computed XOR of the entire packet. This should be 0 if the packet passes.
- **Expected_Checksum** - The XOR included in the CCSDS packet secondary header.

Invalid_Parameter_Info.T:

Record for holding information about an invalid parameter

Table 52: Invalid_Parameter_Info Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Parameter_Types. Parameter_Id	0 to 65535	16	0	15
Errant_Field_ Number	Interfaces. Unsigned_32	0 to 4294967295	32	16	47
Errant_Field	Basic_Types.Poly_ Type	-	64	48	111

Field Descriptions:

- **Id** - The parameter Id received.
- **Errant_Field_Number** - The field that was invalid. 1 is the first field, 0 means unknown field, 2**32 means that the length field of the parameter was invalid.
- **Errant_Field** - A polymorphic type containing the bad field data, or length when Errant_Field_Number is 2**32.

Operands.T:

This is a type that contains a left and right side

Table 53: Operands Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Left	Interfaces. Unsigned_16	0 to 65535	16	0	15
Right	Interfaces. Unsigned_16	0 to 65535	16	16	31

Field Descriptions:

- **Left** - The left side

- **Right** - The right side

Packed_Address.T:

A packed system address.

Table 54: Packed_Address Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Address	System.Address	-	32	0	31

Field Descriptions:
- **Address** - The starting address of the memory region.

Packed_Connector_Index.T:

Single component record for holding packed connector index.

Table 55: Packed_Connector_Index Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Index	Connector_Types. Connector_Index_ Type	1 to 65535	16	0	15

Field Descriptions:
- **Index** - The 16-bit connector index.

Packed_Enable_Disable_Type.T:

Single component record for holding an enable/disable enumeration.

Table 56: Packed_Enable_Disable_Type Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
State	Basic_Enums. Enable_Disable_ Type.E	0 => Disabled 1 => Enabled	8	0	7

Field Descriptions:
- **State** - The 8-bit enable disable enumeration.

Packed_Exception_Occurrence.T:

Packed record which holds information from an Ada Exception Occurrence type. This is the type passed into the Last Chance Handler when running a full runtime. *Preamble (inline Ada definitions):*

```
type Exception_Name_Buffer is new Basic_Types.Byte_Array (0 .. 99)
  with Size => 100 * 8,
       Object_Size => 100 * 8;
type Exception_Message_Buffer is new Basic_Types.Byte_Array (0 .. 299)
  with Size => 300 * 8,
       Object_Size => 300 * 8;
```

Table 57: Packed_Exception_Occurrence Packed Record : 5472 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Exception_Name	Exception_Name_Buffer	-	800	0	799
Exception_Message	Exception_Message_Buffer	-	2400	800	3199
Stack_Trace_Depth	Interfaces.Unsigned_32	0 to 4294967295	32	3200	3231
Stack_Trace	Stack_Trace_Addresses.T	-	2240	3232	5471

Field Descriptions:

- **Exception_Name** - The exception name.
- **Exception_Message** - The exception message.
- **Stack_Trace_Depth** - The depth of the reported stack trace.
- **Stack_Trace** - The stack trace addresses.

Packed_F32.T:

Single component record for holding packed 32-bit floating point number.

Table 58: Packed_F32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Short_Float	-3.40282e+38 to 3.40282e+38	32	0	31

Field Descriptions:

- **Value** - The 32-bit floating point number.

Packed_Fault_Id.T:

A packed record which holds a fault identifier.

Table 59: Packed_Fault_Id Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Fault_Types.Fault_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The fault identifier

Packed_Integer.T:

Single component record for holding packed Integer value.

Table 60: Packed_Integer Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Integer	-2147483648 to 2147483647	32	0	31

Field Descriptions:
- **Value** - The 32-bit Signed Integer.

Packed_Missed_Pet_Limit.T:

Limit value packed for the task watchdog data products

Table 61: Packed_Missed_Pet_Limit Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Limit	Task_Watchdog_ Types.Missed_Pet_ Limit_Type	1 to 65534	16	0	15

Field Descriptions:
- **Limit** - The limit value

Packed_Natural.T:

Single component record for holding packed Natural value.

Table 62: Packed_Natural Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Natural	0 to 2147483647	32	0	31

Field Descriptions:
- **Value** - The 32-bit Natural Integer.

Packed_U16.T:

Single component record for holding packed unsigned 16-bit value.

Table 63: Packed_U16 Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Interfaces. Unsigned_16	0 to 65535	16	0	15

Field Descriptions:
- **Value** - The 16-bit unsigned integer.

Packed_U32.T:

Single component record for holding packed unsigned 32-bit value.

Table 64: Packed_U32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Interfaces. Unsigned_32	0 to 4294967295	32	0	31

Field Descriptions:
- **Value** - The 32-bit unsigned integer.

Packed_Watchdog_Component_State.T:

State value packed for the task watchdog data products

Table 65: Packed_Watchdog_Component_State Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
State	Task_Watchdog_ Enums.Watchdog_ Enabled_State.E	0 => Disabled 1 => Enabled	8	0	7

Field Descriptions:
- **State** - The state of the watchdog component to check all upstream petters.

Packet.T:

Generic packet for holding arbitrary data

Table 66: Packet Packed Record : 10080 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Packet_ Header.T	-	112	0	111	-
Buffer	Packet_ Types.Packet_ Buffer_Type	-	9968	112	10079	Header. Buffer_Length

Field Descriptions:
- **Header** - The packet header
- **Buffer** - A buffer that contains the packet data

Packet_Data_Product_Ids.T:

A packed record which holds a packet identifier and data product identifier.

Table 67: Packet_Data_Product_Ids Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Packet_Id	Packet_Types. Packet_Id	0 to 65535	16	0	15
Data_Product_Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	16	31

Field Descriptions:

- **Packet_Id** - The packet identifier
- **Data_Product_Id** - The data product identifier

Packet_Header.T:

Generic packet header for holding arbitrary data

Table 68: Packet_Header Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	64	0	63
Id	Packet_Types. Packet_Id	0 to 65535	16	64	79
Sequence_Count	Packet_Types. Sequence_Count_Mod_ Type	0 to 16383	16	80	95
Buffer_Length	Packet_Types. Packet_Buffer_ Length_Type	0 to 1246	16	96	111

Field Descriptions:

- **Time** - The timestamp for the packet item.
- **Id** - The packet identifier
- **Sequence_Count** - Packet Sequence Count
- **Buffer_Length** - The number of bytes used in the packet buffer

Packet_Id.T:

A packed record which holds a packet identifier.

Table 69: Packet_Id Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Packet_Types. Packet_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The packet identifier

Packet_Period.T:

A packed record which holds a packet identifier and period.

Table 70: Packet_Period Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Packet_Types. Packet_Id	0 to 65535	16	0	15
Period	Natural	0 to 2147483647	32	16	47

Field Descriptions:

- **Id** - The packet identifier
- **Period** - The packet period, in ticks

Parameter.T:

Generic parameter packet for holding a generic parameter

Table 71: Parameter Packed Record : 280 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Parameter_ Header.T	-	24	0	23	–
Buffer	Parameter_ Types. Parameter_ Buffer_Type	-	256	24	279	Header.Buffer_ Length

Field Descriptions:

- **Header** - The parameter header
- **Buffer** - A buffer that contains the parameter type

Parameter_Header.T:

Generic parameter header for holding arbitrary parameters

Table 72: Parameter_Header Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Parameter_Types. Parameter_Id	0 to 65535	16	0	15
Buffer_Length	Parameter_Types. Parameter_Buffer_ Length_Type	0 to 32	8	16	23

Field Descriptions:

- **Id** - The parameter identifier
- **Buffer_Length** - The number of bytes used in the parameter type buffer

Parameter_Update.T:

A record intended to be used as a provide/modify connector type for updating/fetching parameters.

Table 73: Parameter_Update Packed Record : 312 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Table_Id	Parameter_Types.Parameter_Table_Id	0 to 65535	16	0	15	-
Operation	Parameter_Enums.Parameter_Operation_Type.E	0 => Stage 1 => Update 2 => Fetch 3 => Validate	8	16	23	-
Status	Parameter_Enums.Parameter_Update_Status.E	0 => Success 1 => Id_Error 2 => Validation_Error 3 => Length_Error	8	24	31	-
Param	Parameter.T	-	280	32	311	-

Field Descriptions:

- **`Table_Id`** - The ID for the table that contains this parameter
- **`Operation`** - The parameter operation to perform.
- **`Status`** - The parameter return status.
- **`Param`** - The parameter that has been updated or fetched.

Pet.T:

The pet datatype is used for servicing a watchdog. Included in this type is a count.

Table 74: Pet Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Count	Interfaces.Unsigned_32	0 to 4294967295	32	0	31

Field Descriptions:

- **`Count`** - The cycle number of the pet.

Pico_Example_Cpu_Monitor_Packet_Type.T:

This is the autocoded packet type for the CPU monitor component. It contains CPU utilization information for every task in the assembly.

Table 75: Pico_Example_Cpu_Monitor_Packet_Type Packed Record : 192 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit

Fault_Correction_ Instance_Active_ Usage_1	Basic_Types.Byte	0 to 255	8	0	7
Fault_Correction_ Instance_Active_ Usage_2	Basic_Types.Byte	0 to 255	8	8	15
Fault_Correction_ Instance_Active_ Usage_3	Basic_Types.Byte	0 to 255	8	16	23
Ticker_Instance_ Active_Usage_1	Basic_Types.Byte	0 to 255	8	24	31
Ticker_Instance_ Active_Usage_2	Basic_Types.Byte	0 to 255	8	32	39
Ticker_Instance_ Active_Usage_3	Basic_Types.Byte	0 to 255	8	40	47
Watchdog_Rate_Group_ Active_Usage_1	Basic_Types.Byte	0 to 255	8	48	55
Watchdog_Rate_Group_ Active_Usage_2	Basic_Types.Byte	0 to 255	8	56	63
Watchdog_Rate_Group_ Active_Usage_3	Basic_Types.Byte	0 to 255	8	64	71
Slow_Rate_Group_ Active_Usage_1	Basic_Types.Byte	0 to 255	8	72	79
Slow_Rate_Group_ Active_Usage_2	Basic_Types.Byte	0 to 255	8	80	87
Slow_Rate_Group_ Active_Usage_3	Basic_Types.Byte	0 to 255	8	88	95
Fast_Rate_Group_ Active_Usage_1	Basic_Types.Byte	0 to 255	8	96	103
Fast_Rate_Group_ Active_Usage_2	Basic_Types.Byte	0 to 255	8	104	111
Fast_Rate_Group_ Active_Usage_3	Basic_Types.Byte	0 to 255	8	112	119
Command_Router_ Instance_Active_ Usage_1	Basic_Types.Byte	0 to 255	8	120	127
Command_Router_ Instance_Active_ Usage_2	Basic_Types.Byte	0 to 255	8	128	135
Command_Router_ Instance_Active_ Usage_3	Basic_Types.Byte	0 to 255	8	136	143
Ccsds_Serial_ Interface_Instance_ Active_Usage_1	Basic_Types.Byte	0 to 255	8	144	151
Ccsds_Serial_ Interface_Instance_ Active_Usage_2	Basic_Types.Byte	0 to 255	8	152	159
Ccsds_Serial_ Interface_Instance_ Active_Usage_3	Basic_Types.Byte	0 to 255	8	160	167
Ccsds_Serial_ Interface_Instance_ Listener_Usage_1	Basic_Types.Byte	0 to 255	8	168	175

Ccsds_Serial_ Interface_Instance_ Listener_Usage_2	Basic_Types.Byte	0 to 255	8	176	183
Ccsds_Serial_ Interface_Instance_ Listener_Usage_3	Basic_Types.Byte	0 to 255	8	184	191

Field Descriptions:

- **Fault_Correction_Instance_Active_Usage_1** - The period one CPU utilization percentage for the Fault_Correction_Instance.Active task.
- **Fault_Correction_Instance_Active_Usage_2** - The period two CPU utilization percentage for the Fault_Correction_Instance.Active task.
- **Fault_Correction_Instance_Active_Usage_3** - The period three CPU utilization percentage for the Fault_Correction_Instance.Active task.
- **Ticker_Instance_Active_Usage_1** - The period one CPU utilization percentage for the Ticker_Instance.Active task.
- **Ticker_Instance_Active_Usage_2** - The period two CPU utilization percentage for the Ticker_Instance.Active task.
- **Ticker_Instance_Active_Usage_3** - The period three CPU utilization percentage for the Ticker_Instance.Active task.
- **Watchdog_Rate_Group_Active_Usage_1** - The period one CPU utilization percentage for the Watchdog_Rate_Group.Active task.
- **Watchdog_Rate_Group_Active_Usage_2** - The period two CPU utilization percentage for the Watchdog_Rate_Group.Active task.
- **Watchdog_Rate_Group_Active_Usage_3** - The period three CPU utilization percentage for the Watchdog_Rate_Group.Active task.
- **Slow_Rate_Group_Active_Usage_1** - The period one CPU utilization percentage for the Slow_Rate_Group.Active task.
- **Slow_Rate_Group_Active_Usage_2** - The period two CPU utilization percentage for the Slow_Rate_Group.Active task.
- **Slow_Rate_Group_Active_Usage_3** - The period three CPU utilization percentage for the Slow_Rate_Group.Active task.
- **Fast_Rate_Group_Active_Usage_1** - The period one CPU utilization percentage for the Fast_Rate_Group.Active task.
- **Fast_Rate_Group_Active_Usage_2** - The period two CPU utilization percentage for the Fast_Rate_Group.Active task.
- **Fast_Rate_Group_Active_Usage_3** - The period three CPU utilization percentage for the Fast_Rate_Group.Active task.
- **Command_Router_Instance_Active_Usage_1** - The period one CPU utilization percentage for the Command_Router_Instance.Active task.
- **Command_Router_Instance_Active_Usage_2** - The period two CPU utilization percentage for the Command_Router_Instance.Active task.
- **Command_Router_Instance_Active_Usage_3** - The period three CPU utilization percentage for the Command_Router_Instance.Active task.
- **Ccsds_Serial_Interface_Instance_Active_Usage_1** - The period one CPU utilization percentage for the Ccsds_Serial_Interface_Instance.Active task.
- **Ccsds_Serial_Interface_Instance_Active_Usage_2** - The period two CPU utiliza-

tion percentage for the Ccsds_Serial_Interface_Instance.Active task.

- **`Ccsds_Serial_Interface_Instance_Active_Usage_3`** - The period three CPU utilization percentage for the Ccsds_Serial_Interface_Instance.Active task.
- **`Ccsds_Serial_Interface_Instance_Listener_Usage_1`** - The period one CPU utilization percentage for the Ccsds_Serial_Interface_Instance.Listener task.
- **`Ccsds_Serial_Interface_Instance_Listener_Usage_2`** - The period two CPU utilization percentage for the Ccsds_Serial_Interface_Instance.Listener task.
- **`Ccsds_Serial_Interface_Instance_Listener_Usage_3`** - The period three CPU utilization percentage for the Ccsds_Serial_Interface_Instance.Listener task.

Pico_Example_Fault_Responses_Packed_Id_Type.T:

This is an autocoded packed ID type for the fault correction component based on the fault IDs that it handles.

Table 76: Pico_Example_Fault_Responses_Packed_Id_Type Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Pico_Example_Fault_Responses_Enums.Fault_Type.E	0 => None 1 => Task_Watchdog_Instance_Slow_Rate_Group_Fault 2 => Task_Watchdog_Instance_Fast_Rate_Group_Fault 3 => Fault_Producer_Instance_Fault_1 4 => Fault_Producer_Instance_Fault_2	16	0	15

Field Descriptions:

- **`Id`** - The fault type identifier.

Pico_Example_Fault_Responses_Status_Record.T:

This is an autocoded data product status record type for a Fault Correction component.

Table 77: Pico_Example_Fault_Responses_Status_Record Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Task_Watchdog_Instance_Slow_Rate_Group_Fault_Status	Fault_Correction_Enums.Status_Type.E	0 => Disabled 1 => Nominal 2 => Fault_Detected 3 => Fault_Latched	2	0	1
Task_Watchdog_Instance_Fast_Rate_Group_Fault_Status	Fault_Correction_Enums.Status_Type.E	0 => Disabled 1 => Nominal 2 => Fault_Detected 3 => Fault_Latched	2	2	3

Fault_Producer_ Instance_Fault_ 1_Status	Fault_ Correction_ Enums.Status_ Type.E	0 => Disabled 1 => Nominal 2 => Fault_Detected 3 => Fault_Latched	2	4	5
Fault_Producer_ Instance_Fault_ 2_Status	Fault_ Correction_ Enums.Status_ Type.E	0 => Disabled 1 => Nominal 2 => Fault_Detected 3 => Fault_Latched	2	6	7

Field Descriptions:

- **Task_Watchdog_Instance_Slow_Rate_Group_Fault_Status** - The current status of the Task_Watchdog_Instance.Slow_Rate_Group_Fault fault response.
- **Task_Watchdog_Instance_Fast_Rate_Group_Fault_Status** - The current status of the Task_Watchdog_Instance.Fast_Rate_Group_Fault fault response.
- **Fault_Producer_Instance_Fault_1_Status** - The current status of the Fault_Producer_Instance.Fault_1 fault response.
- **Fault_Producer_Instance_Fault_2_Status** - The current status of the Fault_Producer_Instance.Fault_2 fault response.

Pico_Example_Queue_Monitor_Packet_Type.T:

This is the autocoded packet type for the Queue Monitor component. It contains queue utilization information for each queued component in the assembly.

Table 78: Pico_Example_Queue_Monitor_Packet_Type Packed Record : 160 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Slow_Rate_Group_ Current_Usage	Basic_Types.Byte	0 to 255	8	0	7
Slow_Rate_Group_ Maximum_Usage	Basic_Types.Byte	0 to 255	8	8	15
Fast_Rate_Group_ Current_Usage	Basic_Types.Byte	0 to 255	8	16	23
Fast_Rate_Group_ Maximum_Usage	Basic_Types.Byte	0 to 255	8	24	31
Watchdog_Rate_Group_ Current_Usage	Basic_Types.Byte	0 to 255	8	32	39
Watchdog_Rate_Group_ Maximum_Usage	Basic_Types.Byte	0 to 255	8	40	47
Command_Router_ Instance_Current_ Usage	Basic_Types.Byte	0 to 255	8	48	55
Command_Router_ Instance_Maximum_ Usage	Basic_Types.Byte	0 to 255	8	56	63
Product_Packetizer_ Instance_Current_ Usage	Basic_Types.Byte	0 to 255	8	64	71

Product_Packetizer_ Instance_Maximum_ Usage	Basic_Types.Byte	0 to 255	8	72	79
Ccsds_Serial_ Interface_Instance_ Current_Usage	Basic_Types.Byte	0 to 255	8	80	87
Ccsds_Serial_ Interface_Instance_ Maximum_Usage	Basic_Types.Byte	0 to 255	8	88	95
Counter_Instance_ Current_Usage	Basic_Types.Byte	0 to 255	8	96	103
Counter_Instance_ Maximum_Usage	Basic_Types.Byte	0 to 255	8	104	111
Oscillator_A_ Current_Usage	Basic_Types.Byte	0 to 255	8	112	119
Oscillator_A_ Maximum_Usage	Basic_Types.Byte	0 to 255	8	120	127
Oscillator_B_ Current_Usage	Basic_Types.Byte	0 to 255	8	128	135
Oscillator_B_ Maximum_Usage	Basic_Types.Byte	0 to 255	8	136	143
Fault_Correction_ Instance_Current_ Usage	Basic_Types.Byte	0 to 255	8	144	151
Fault_Correction_ Instance_Maximum_ Usage	Basic_Types.Byte	0 to 255	8	152	159

Field Descriptions:

- **Slow_Rate_Group_Current_Usage** - The current percent usage of the Slow_Rate_Group internal queue.
- **Slow_Rate_Group_Maximum_Usage** - The maximum percent usage (high water mark) of the Slow_Rate_Group internal queue.
- **Fast_Rate_Group_Current_Usage** - The current percent usage of the Fast_Rate_Group internal queue.
- **Fast_Rate_Group_Maximum_Usage** - The maximum percent usage (high water mark) of the Fast_Rate_Group internal queue.
- **Watchdog_Rate_Group_Current_Usage** - The current percent usage of the Watchdog_Rate_Group internal queue.
- **Watchdog_Rate_Group_Maximum_Usage** - The maximum percent usage (high water mark) of the Watchdog_Rate_Group internal queue.
- **Command_Router_Instance_Current_Usage** - The current percent usage of the Command_Router_Instance internal queue.
- **Command_Router_Instance_Maximum_Usage** - The maximum percent usage (high water mark) of the Command_Router_Instance internal queue.
- **Product_Packetizer_Instance_Current_Usage** - The current percent usage of the Product_Packetizer_Instance internal queue.
- **Product_Packetizer_Instance_Maximum_Usage** - The maximum percent usage (high water mark) of the Product_Packetizer_Instance internal queue.
- **Ccsds_Serial_Interface_Instance_Current_Usage** - The current percent usage of the Ccsds_Serial_Interface_Instance internal queue.

- **Ccsds_Serial_Interface_Instance_Maximum_Usage** - The maximum percent usage (high water mark) of the Ccsds_Serial_Interface_Instance internal queue.
- **Counter_Instance_Current_Usage** - The current percent usage of the Counter_Instance internal queue.
- **Counter_Instance_Maximum_Usage** - The maximum percent usage (high water mark) of the Counter_Instance internal queue.
- **Oscillator_A_Current_Usage** - The current percent usage of the Oscillator_A internal queue.
- **Oscillator_A_Maximum_Usage** - The maximum percent usage (high water mark) of the Oscillator_A internal queue.
- **Oscillator_B_Current_Usage** - The current percent usage of the Oscillator_B internal queue.
- **Oscillator_B_Maximum_Usage** - The maximum percent usage (high water mark) of the Oscillator_B internal queue.
- **Fault_Correction_Instance_Current_Usage** - The current percent usage of the Fault_Correction_Instance internal queue.
- **Fault_Correction_Instance_Maximum_Usage** - The maximum percent usage (high water mark) of the Fault_Correction_Instance internal queue.

Pico_Example_Stack_Monitor_Packet_Type.T:

This is the autocoded packet type for the stack monitor component. It contains stack and secondary stack utilization information for every task in the assembly.

Table 79: Pico_Example_Stack_Monitor_Packet_Type Packed Record : 128 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Fault_Correction_Instance_Active_Primary_Stack_Usage	Basic_Types.Byte	0 to 255	8	0	7
Fault_Correction_Instance_Active_Secondary_Stack_Usage	Basic_Types.Byte	0 to 255	8	8	15
Ticker_Instance_Active_Primary_Stack_Usage	Basic_Types.Byte	0 to 255	8	16	23
Ticker_Instance_Active_Secondary_Stack_Usage	Basic_Types.Byte	0 to 255	8	24	31
Watchdog_Rate_Group_Active_Primary_Stack_Usage	Basic_Types.Byte	0 to 255	8	32	39
Watchdog_Rate_Group_Active_Secondary_Stack_Usage	Basic_Types.Byte	0 to 255	8	40	47
Slow_Rate_Group_Active_Primary_Stack_Usage	Basic_Types.Byte	0 to 255	8	48	55
Slow_Rate_Group_Active_Secondary_Stack_Usage	Basic_Types.Byte	0 to 255	8	56	63

Fast_Rate_Group_ Active_Primary_ Stack_Usage	Basic_Types.Byte	0 to 255	8	64	71
Fast_Rate_Group_ Active_Secondary_ Stack_Usage	Basic_Types.Byte	0 to 255	8	72	79
Command_Router_ Instance_Active_ Primary_Stack_Usage	Basic_Types.Byte	0 to 255	8	80	87
Command_Router_ Instance_Active_ Secondary_Stack_ Usage	Basic_Types.Byte	0 to 255	8	88	95
Ccsds_Serial_ Interface_Instance_ Active_Primary_ Stack_Usage	Basic_Types.Byte	0 to 255	8	96	103
Ccsds_Serial_ Interface_Instance_ Active_Secondary_ Stack_Usage	Basic_Types.Byte	0 to 255	8	104	111
Ccsds_Serial_ Interface_Instance_ Listener_Primary_ Stack_Usage	Basic_Types.Byte	0 to 255	8	112	119
Ccsds_Serial_ Interface_Instance_ Listener_Secondary_ Stack_Usage	Basic_Types.Byte	0 to 255	8	120	127

Field Descriptions:

- **Fault_Correction_Instance_Active_Primary_Stack_Usage** - The primary stack utilization percentage for the Fault_Correction_Instance.Active task.
- **Fault_Correction_Instance_Active_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Fault_Correction_Instance.Active task.
- **Ticker_Instance_Active_Primary_Stack_Usage** - The primary stack utilization percentage for the Ticker_Instance.Active task.
- **Ticker_Instance_Active_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Ticker_Instance.Active task.
- **Watchdog_Rate_Group_Active_Primary_Stack_Usage** - The primary stack utilization percentage for the Watchdog_Rate_Group.Active task.
- **Watchdog_Rate_Group_Active_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Watchdog_Rate_Group.Active task.
- **Slow_Rate_Group_Active_Primary_Stack_Usage** - The primary stack utilization percentage for the Slow_Rate_Group.Active task.
- **Slow_Rate_Group_Active_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Slow_Rate_Group.Active task.
- **Fast_Rate_Group_Active_Primary_Stack_Usage** - The primary stack utilization percentage for the Fast_Rate_Group.Active task.
- **Fast_Rate_Group_Active_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Fast_Rate_Group.Active task.

- **Command_Router_Instance_Active_Primary_Stack_Usage** - The primary stack utilization percentage for the Command_Router_Instance.Active task.
- **Command_Router_Instance_Active_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Command_Router_Instance.Active task.
- **Ccsds_Serial_Interface_Instance_Active_Primary_Stack_Usage** - The primary stack utilization percentage for the Ccsds_Serial_Interface_Instance.Active task.
- **Ccsds_Serial_Interface_Instance_Active_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Ccsds_Serial_Interface_Instance.Active task.
- **Ccsds_Serial_Interface_Instance_Listener_Primary_Stack_Usage** - The primary stack utilization percentage for the Ccsds_Serial_Interface_Instance.Listener task.
- **Ccsds_Serial_Interface_Instance_Listener_Secondary_Stack_Usage** - The secondary stack utilization percentage for the Ccsds_Serial_Interface_Instance.Listener task.

Pico_Example_Task_Watchdog_List_State_Record.T:

This is an autocoded data product status record type for a Task Watchdog component regarding the status of checking each connector for pets.

Table 80: Pico_Example_Task_Watchdog_List_State_Record Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Slow_Rate_Group_State	Task_Watchdog_Enums.Watchdog_Action_State.E	0 => Disabled 1 => Warn 2 => Error_Fault	2	0	1
Fast_Rate_Group_State	Task_Watchdog_Enums.Watchdog_Action_State.E	0 => Disabled 1 => Warn 2 => Error_Fault	2	2	3
Reserved_0	Task_Watchdog_Types.Two_Bit_Padding_Type	0 to 3	2	4	5
Reserved_1	Task_Watchdog_Types.Two_Bit_Padding_Type	0 to 3	2	6	7

Field Descriptions:

- **Slow_Rate_Group_State** - The current state of the Slow_Rate_Group watchdog action.
- **Fast_Rate_Group_State** - The current state of the Fast_Rate_Group watchdog action.
- **Reserved_0** - Padding bits, not used.
- **Reserved_1** - Padding bits, not used.

Pico_Example_Task_Watchdog_List_Watchdog_Action_Cmd.T:

This is an autocoded packed enumeration record which contains an enumeration literal for each task that the task watchdog manages. This record contains information for changing the limit of a specific watchdog connector via this enumeration. *Preamble (inline Ada definitions):*

```
type Task_Enum_Type is (
  Slow_Rate_Group,
```

```
  Fast_Rate_Group
);
for Task_Enum_Type use (
  Slow_Rate_Group => 1,
  Fast_Rate_Group => 2
);
```

Table 81: Pico_Example_Task_Watchdog_List_Watchdog_Action_Cmd Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Watchdog_Task	Task_Enum_Type	1 => Slow_Rate_Group 2 => Fast_Rate_Group	16	0	15
New_Action	Task_Watchdog_Enums.Watchdog_Action_State.E	0 => Disabled 1 => Warn 2 => Error_Fault	8	16	23

Field Descriptions:

- **Watchdog_Task** - The task watchdog task enumeration.
- **New_Action** - The new value of the action for the specific associated connector

Pico_Example_Task_Watchdog_List_Watchdog_Limit_Cmd.T:

This is an autocoded packed enumeration record which contains an enumeration literal for each task that the task watchdog manages. This record contains information for changing the limit of a specific watchdog connector via this enumeration. *Preamble (inline Ada definitions):*

```
type Task_Enum_Type is (
  Slow_Rate_Group,
  Fast_Rate_Group
);
for Task_Enum_Type use (
  Slow_Rate_Group => 1,
  Fast_Rate_Group => 2
);
```

Table 82: Pico_Example_Task_Watchdog_List_Watchdog_Limit_Cmd Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Watchdog_Task	Task_Enum_Type	1 => Slow_Rate_Group 2 => Fast_Rate_Group	16	0	15
New_Limit	Task_Watchdog_Types.Missed_Pet_Limit_Type	1 to 65534	16	16	31

Field Descriptions:

- **Watchdog_Task** - The task watchdog task enumeration.
- **New_Limit** - The new value of the limit for the specific associated connector

Stack_Trace_Addresses.T:

An array of packed addresses in big endian. This is sized to easily fit a normal stack trace.

Table 83: Stack_Trace_Addresses Packed Array : 2240 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Packed_Address.T	-	32	70	2240

Sys_Time.T:

A record which holds a time stamp using GPS format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 84: Sys_Time Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces. Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces. Unsigned_32	0 to 4294967295	32	32	63

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of 1/(2^32) sub-seconds.

Task_Timing_Report.T:

Record which holds timing reports for the all-time maximum and recent maximum of an executing task

Table 85: Task_Timing_Report Packed Record : 256 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Max	Timing_Report.T	-	128	0	127
Recent_Max	Timing_Report.T	-	128	128	255

Field Descriptions:

- **Max** - The maximum recorded timing report since start up.
- **Recent_Max** - The maximum recorded timing report during some recent time interval.

Td_Full_Queue_Param.T:

This is a type that contains useful information about a component full queue.

Table 86: Td_Full_Queue_Param Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Dropped_Tick	Tick.T	-	96	0	95

Index	Connector_Types. Connector_Index_ Type	1 to 65535	16	96	111

Field Descriptions:

- **Dropped_Tick** - The tick during which the component's queue was found to be full.
- **Index** - The tick divider index number of the component that had a full queue.

Tick.T:

The tick datatype used for periodic scheduling. Included in this type is the Time associated with a tick and a count.

Table 87: Tick Packed Record : 96 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	64	0	63
Count	Interfaces. Unsigned_32	0 to 4294967295	32	64	95

Field Descriptions:

- **Time** - The timestamp associated with the tick.
- **Count** - The cycle number of the tick.

Time_Exceeded.T:

Datatype used for reporting time deltas at a particular count number.

Table 88: Time_Exceeded Packed Record : 96 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time_Delta	Delta_Time.T	-	64	0	63
Count	Interfaces. Unsigned_32	0 to 4294967295	32	64	95

Field Descriptions:

- **Time_Delta** - The time delta.
- **Count** - The cycle number of the tick.

Timing_Report.T:

Record which holds wall time and execution time to describe the runtime performance of some piece of code.

Table 89: Timing_Report Packed Record : 128 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Wall_Time	Delta_Time.T	-	64	0	63
Execution_Time	Delta_Time.T	-	64	64	127

Field Descriptions:

- **`Wall_Time`** - The wall time associated with the measurement.
- **`Execution_Time`** - The time the task spent executing on the CPU.

Watchdog_Action_Cmd.T:

This record contains information for changing the limit of a specific watchdog connector.

Table 90: Watchdog_Action_Cmd Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Index	Connector_Types. Connector_Index_ Type	1 to 65535	16	0	15
New_Action	Task_Watchdog_ Enums.Watchdog_ Action_State.E	0 => Disabled 1 => Warn 2 => Error_Fault	8	16	23

Field Descriptions:

- **`Index`** - The index of the connector that the command wants to change the limit of
- **`New_Action`** - The new value of the action for the specific associated connector

Watchdog_Limit_Cmd.T:

This record contains information for changing the limit of a specific watchdog connector.

Table 91: Watchdog_Limit_Cmd Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Index	Connector_Types. Connector_Index_ Type	1 to 65535	16	0	15
New_Limit	Task_Watchdog_ Types.Missed_Pet_ Limit_Type	1 to 65534	16	16	31

Field Descriptions:

- **`Index`** - The index of the connector that the command wants to change the limit of
- **`New_Limit`** - The new value of the limit for the specific associated connector

3.3 Enumerations

The following section outlines any enumerations used in the assembly.

Basic_Enums.Enable_Disable_Type.E:

This enumeration includes enable and disable state.

Table 92: Enable_Disable_Type Literals:

Name	Value	Description
Disabled	0	The state is disabled.
Enabled	1	The state is enabled.

Ccsds_Enums.Ccsds_Packet_Type.E:

This single bit is used to identify that this is a Telecommand Packet or a Telemetry Packet. A Telemetry Packet has this bit set to value 0; therefore, for all Telecommand Packets Bit 3 shall be set to value 1.

Table 93: Ccsds_Packet_Type Literals:

Name	Value	Description
Telemetry	0	Indicates a telemetry packet
Telecommand	1	Indicates a telecommand packet

Ccsds_Enums.Ccsds_Secondary_Header_Indicator.E:

This one bit flag signals the presence (Bit 4 = 1) or absence (Bit 4 = 0) of a Secondary Header data structure within the packet.

Table 94: Ccsds_Secondary_Header_Indicator Literals:

Name	Value	Description
Secondary_Header_Not_Present	0	Indicates that the secondary header is not present within the packet
Secondary_Header_Present	1	Indicates that the secondary header is present within the packet

Ccsds_Enums.Ccsds_Sequence_Flag.E:

This flag provides a method for defining whether this packet is a first, last, or intermediate component of a higher layer data structure.

Table 95: Ccsds_Sequence_Flag Literals:

Name	Value	Description
Continuationsegment	0	Continuation component of higher data structure
Firstsegment	1	First component of higher data structure
Lastsegment	2	Last component of higher data structure
Unsegmented	3	Standalone packet

Command_Enums.Command_Response_Status.E:

This status enumeration provides information on the success/failure of a command through the command response connector.

Table 96: Command_Response_Status Literals:

Name	Value	Description

Success	0	Command was passed to the handler and successfully executed.
Failure	1	Command was passed to the handler not successfully executed.
Id_Error	2	Command id was not valid.
Validation_Error	3	Command parameters were not successfully validated.
Length_Error	4	Command length was not correct.
Dropped	5	Command overflowed a component queue and was dropped.
Register	6	This status is used to register a command with the command routing system.
Register_Source	7	This status is used to register command sender's source id with the command router for command response forwarding.

Data_Product_Enums.Fetch_Status.E:

This status denotes whether a data product fetch was successful.

Table 97: Fetch_Status Literals:

Name	Value	Description
Success	0	The data product was returned successfully.
Not_Available	1	No data product is yet available for the provided id.
Id_Out_Of_Range	2	The data product id was out of range.

Event_Filter_Entry_Enums.Global_Filter_State.E:

This is an enumeration indicating if the filter component is enabled/disabled

Table 98: Global_Filter_State Literals:

Name	Value	Description
Disabled	0	Individual event states will be ignored and no events will be filtered
Enabled	1	Individual event states will be used to determine if the event needs to be filtered

Event_Filter_Enums.Issue_Packet_Type.E:

An enumeration for commands once the state is change for ground to determine if they want to send the state packet or not

Table 99: Issue_Packet_Type Literals:

Name	Value	Description
No_Issue	0	Packet will not be issued
Issue	1	Packet will be issued

Event_Limiter_Enums.Issue_Packet_Type.E:

An enumeration for commands once the state is change for ground to determine if they want to send the state packet or not

Table 100: Issue_Packet_Type Literals:

Name	Value	Description
No_Issue	0	Packet will not be issued
Issue	1	Packet will be issued

Fault_Correction_Enums.Status_Type.E:

This status enumerations provides the status of a fault response.

Table 101: Status_Type Literals:

Name	Value	Description
Disabled	0	The fault response is disabled.
Nominal	1	The fault response is enabled and no fault has been detected.
Fault_Detected	2	A fault has been detected and a command response has been issued.
Fault_Latched	3	A fault has been detected and a command response has been issued. The fault has been latched, so any more faults received with this ID will not issue another response until the this fault response has been unlatched.

Parameter_Enums.Parameter_Operation_Type.E:

This enumeration lists the different parameter operations that can be performed.

Table 102: Parameter_Operation_Type Literals:

Name	Value	Description
Stage	0	Stage the parameter.
Update	1	All parameters are staged, it is ok to update all parameters now.
Fetch	2	Fetch the parameter.
Validate	3	Validate all the parameters.

Parameter_Enums.Parameter_Update_Status.E:

This status enumeration provides information on the success/failure of a parameter operation.

Table 103: Parameter_Update_Status Literals:

Name	Value	Description
Success	0	Parameter was successfully staged.
Id_Error	1	Parameter id was not valid.
Validation_Error	2	Parameter values were not successfully validated.
Length_Error	3	Parameter length was not correct.

Pico_Example_Fault_Responses_Enums.Fault_Type.E:

This enumerations lists all the possible fault conditions possible in the assembly. The value of the enumeration corresponds to the fault identifier for that fault.

Table 104: Fault_Type Literals:

Name	Value	Description
None	0	No fault present.
Task_Watchdog_Instance_Slow_Rate_Group_Fault	1	Throw noop with value 1 if this fault occurs.
Task_Watchdog_Instance_Fast_Rate_Group_Fault	2	Throw noop with value 2 if this fault occurs.
Fault_Producer_Instance_Fault_1	3	Throw noop with value 3 if this fault occurs.
Fault_Producer_Instance_Fault_2	4	Throw noop with value 4 if this fault occurs.

Task_Watchdog_Enums.Watchdog_Enabled_State.E:

The state for if each watchdog is enabled or disabled for checking

Table 105: Watchdog_Enabled_State Literals:

Name	Value	Description
Disabled	0	
Enabled	1	

Task_Watchdog_Enums.Watchdog_Action_State.E:

The state for if each watchdog for which action it should take.

Table 106: Watchdog_Action_State Literals:

Name	Value	Description
Disabled	0	
Warn	1	
Error_Fault	2	

Two_Counter_Entry_Enums.Event_State_Type.E:

This is a single bit identifying if the event limiter is enabled for a particular ID.

Table 107: Event_State_Type Literals:

Name	Value	Description
Disabled	0	Event will not be limited
Enabled	1	Event will be limited